# RB4107 ES1 rescue carrier, 1:5 prototype

Track segments: complete pack  (Nazim, 15 Oct 2026)

**PART A: current plan, my track on Nhien's Rev B frame**

1 How it all works: 3D renders, assembled, bent on the stairs, exploded

2 Parts catalogue, one page per part with a highlighted render and vendor link (links clickable)

3 Procurement list: mine vs Nhien's, links, prices, delivery (links clickable)

4 Layout with Nhien's plates: where everything sits, what changed

5 Drawing TS-01 Track (the part I print) and its explainer

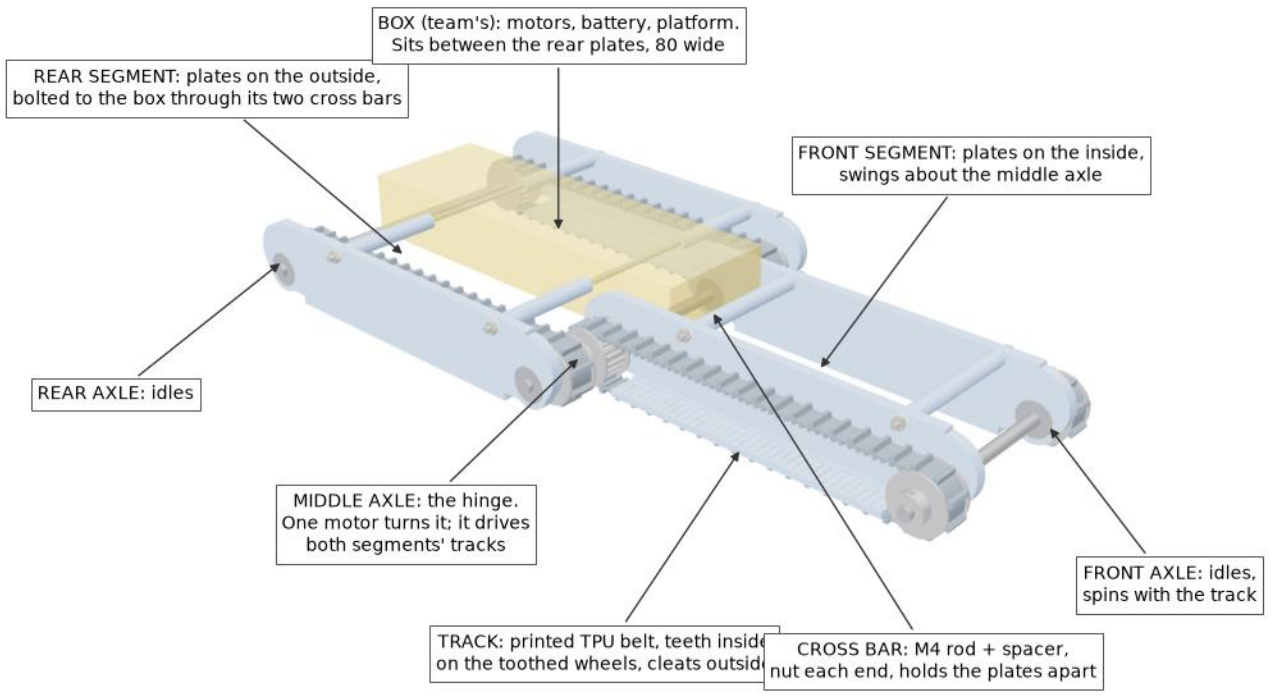
6 Drawing TS-06 Toothed wheel (bought, then bored at the workshop)

**PART B: fallback, the all-printed frame (if the aluminium plates don't happen)**

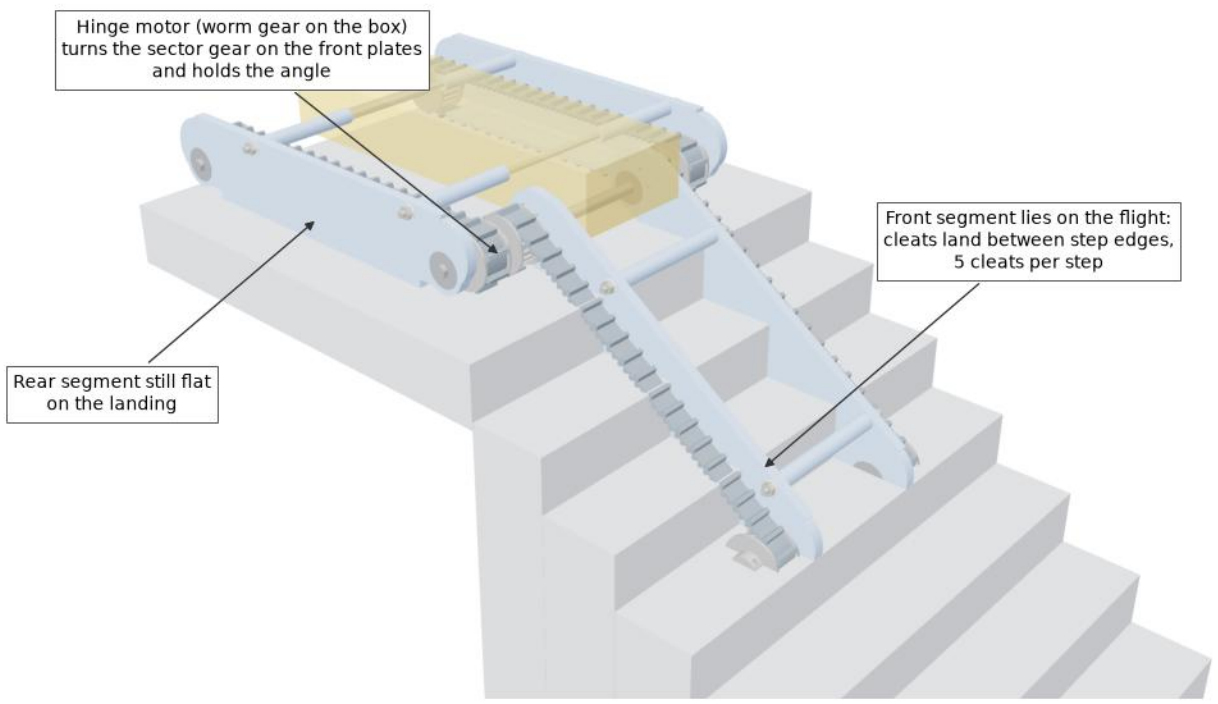
7 Drawings TS-00 to TS-08 with explainers, parts list and build guide, buy list

Files on the branch: track_print/stl (print files), track_print/step (CAD, STEP), track_print/assembly (this pack and its sources).

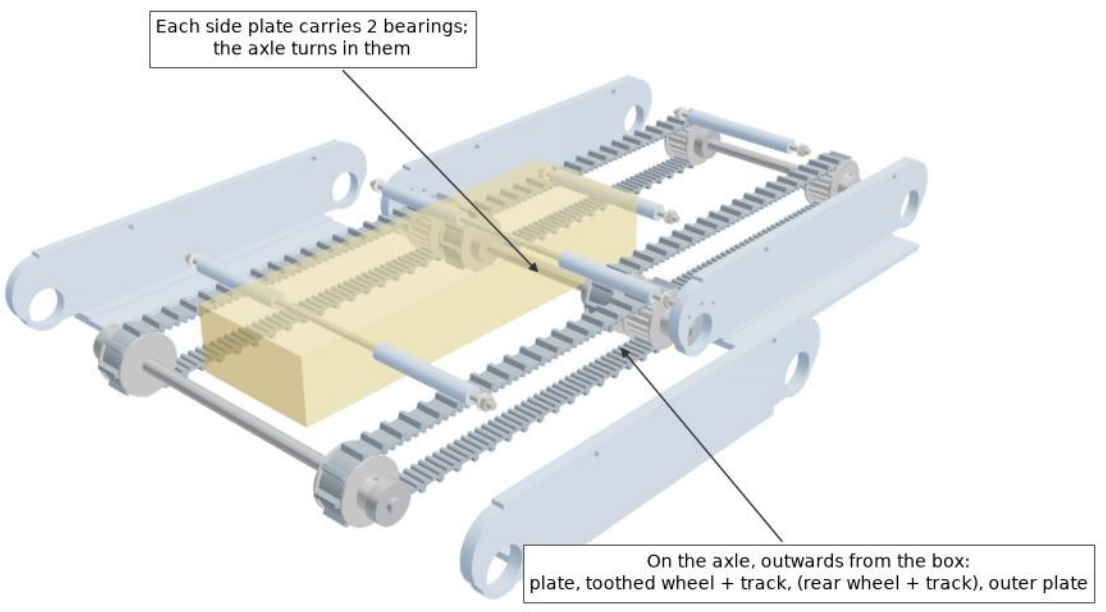
1. Assembled, on the flat




2. Going down the ES1 stairs: front segment bent 35° down over the edge

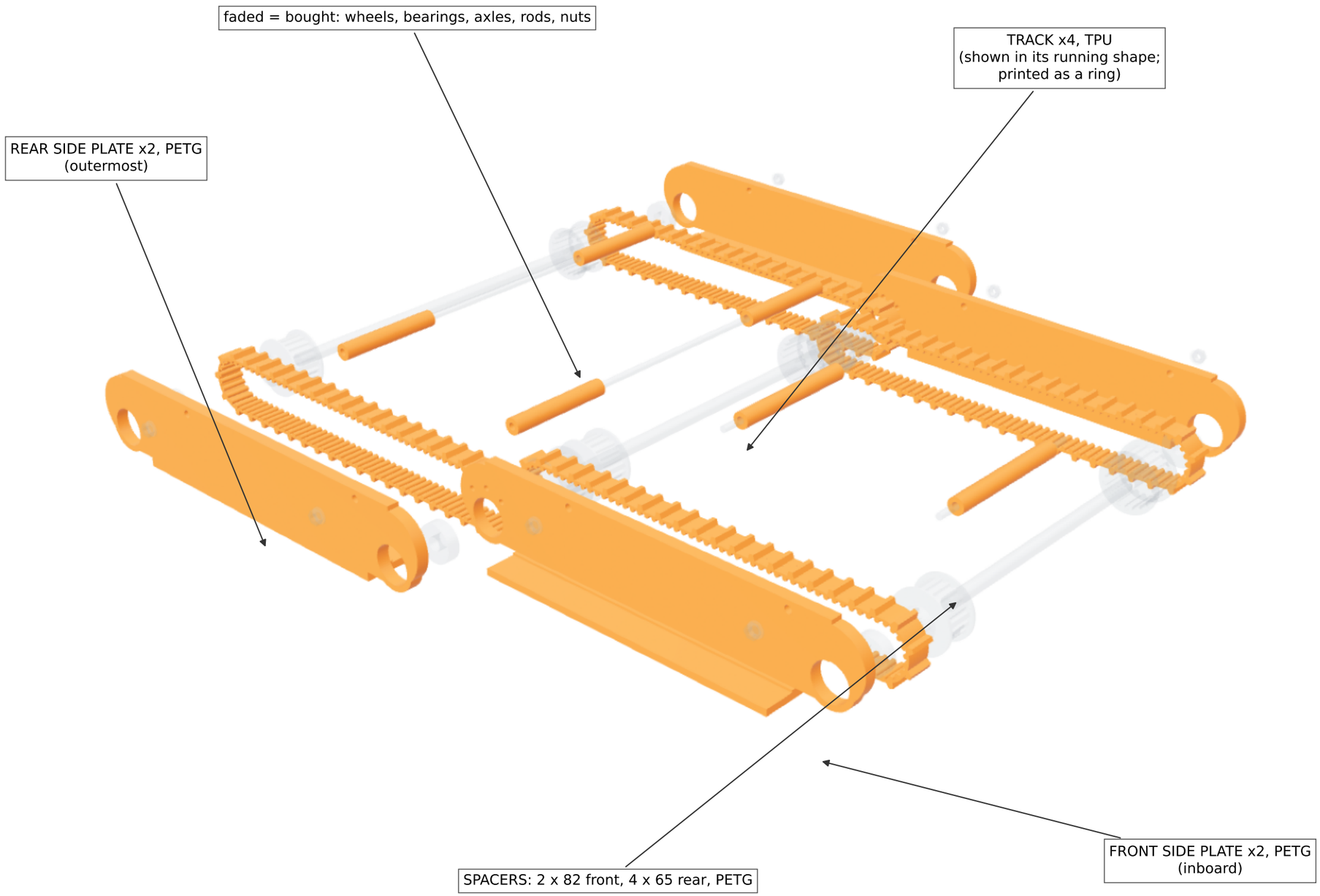



3. Plates pulled out 70 mm each side: what sits on the axles



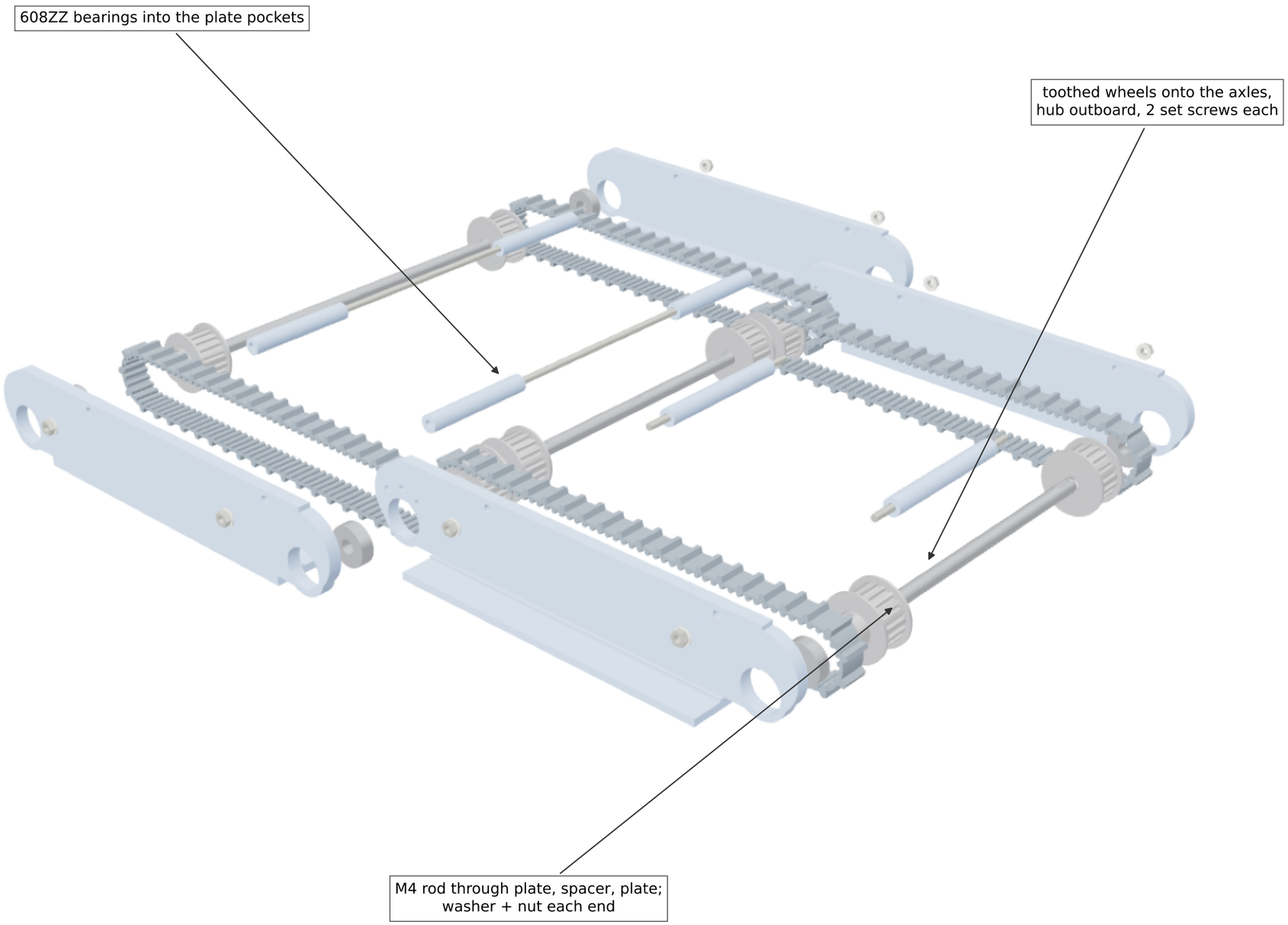

Colours: blue = printed PETG, grey = printed TPU tracks, silver = bought pulleys, bearings, rod. Yellow = box, reference. 1:5 model.

## Exploded view: the parts you print (orange), pulled apart along the axles



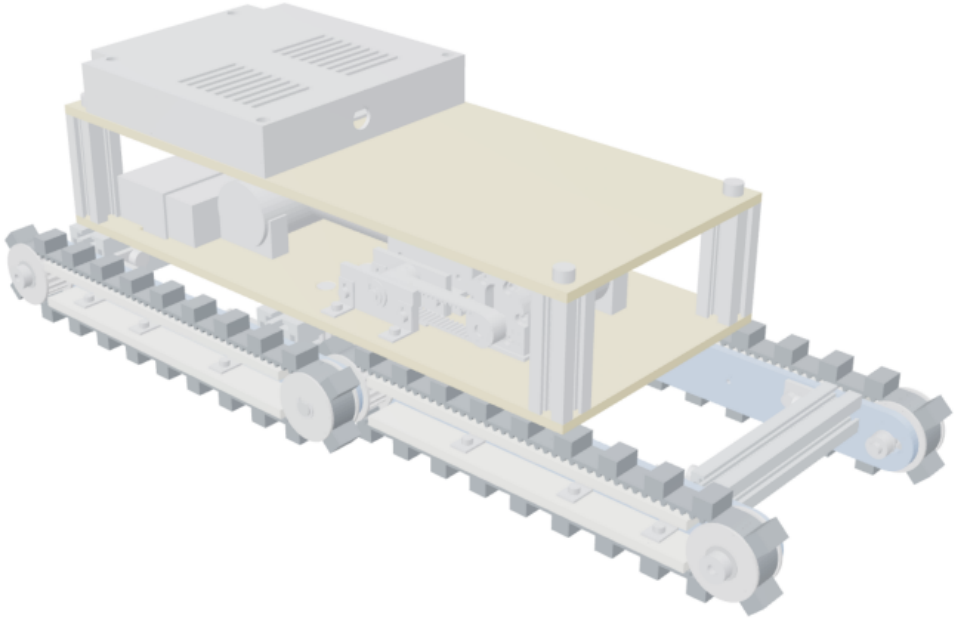

Printed parts: 4 tracks, 4 plates, 6 spacers. Settings and files on the sheets that follow. Rear box / profile frame omitted for clarity.

## Exploded view: everything, same spread




Assembly order: bearings into plates; wheels onto axles; one plate on; track over its two wheels; second plate on; cross bars; nuts.

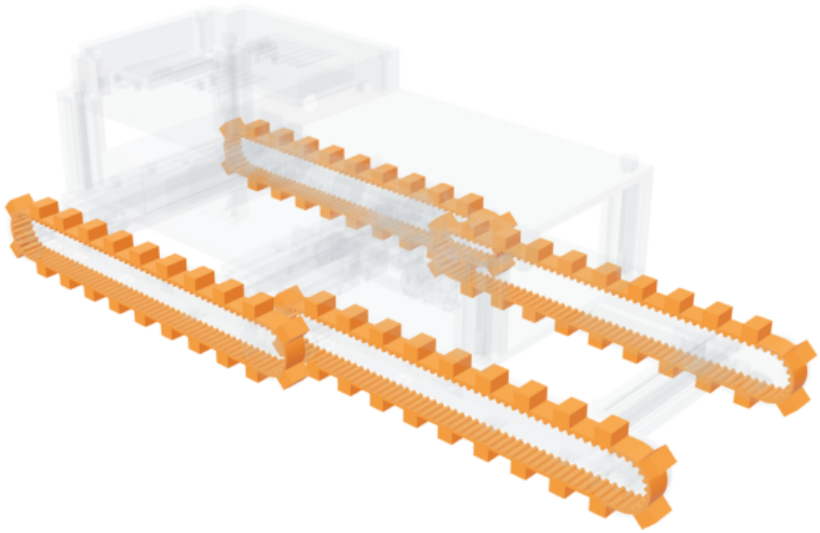
# Parts catalogue, hybrid: my track on Nhien's Rev B frame



One page per part. Orange = that part in Nhien's Rev B assembly (their CAD, 347 parts). Blue-ish parts are links: click the vendor line to open the product page. Rev B sheet numbers refer to 'RB4107 Fabrication drawings (Rev B).pdf'.

Mine to make: track x4.
Mine to buy: pulleys x8, TPU.
Nhien's: plates, rails, shafts, bearings, shoulder screws, bars, deck, motors, electronics (their BOM).

# Track



Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Quantity: 4

**MAKE: 3D print (mine)**

## What it does

The printed TPU belt: 110 teeth inside on the toothed wheels, 45 cleats outside matched to the ES1 step. Unchanged by the plate swap; fits Rev B's pulleys and 225 mm spacing. Rev B's own alternative is a stock 550-5M-15 belt with 22 glued cleats.

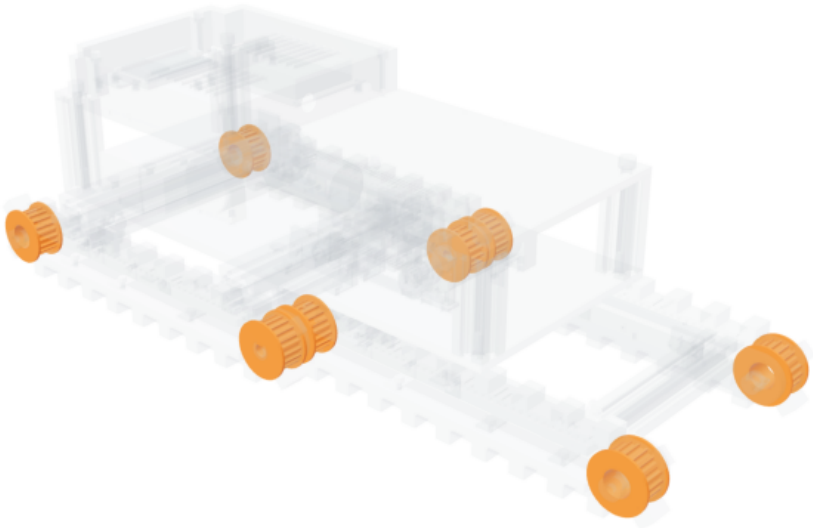
## Where to get it

Fab Lab E4-02-06, Tiertime UP300 with the TPU head. File stl/track_TPU_x4.stl, standing as a ring, no supports, 0.2 mm, 100% infill. Filament: Fab Lab stock or Element14 SG TPU 95A (link).

Link: Element14 SG: TPU 95A filament

https://sg.element14.com/multicomp-pro/mc011460/3d-printer-filament-tpu-1-75

# Toothed wheel (timing pulley) HTD 5M 20T



Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Quantity: 8

**BUY plain bore + MODIFY**

## What it does

Same wheel as before, but Rev B's lanes are only 19-20 mm wide, so there is no room for a set-screw hub. The hinge ones are driven by the D-shaped shaft; the end ones spin on bearings on a shoulder screw.

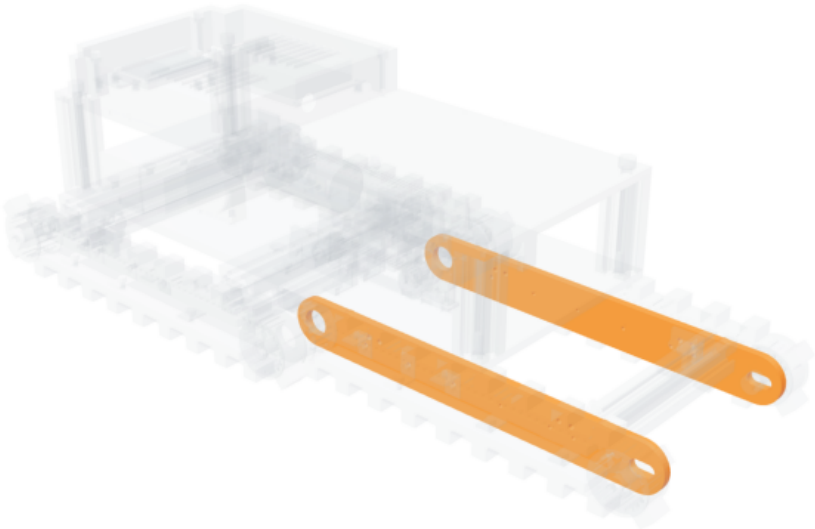
## Where to get it

MISUMI SG, Timing Pulleys HTD 5M (link): 20 T, 15 mm belt, flanges both sides, PILOT (plain) bore. Then Professional Workshop: 4 bored to a D-shape Ø8 (hinge shaft, M1), 4 bored to Ø16 H7 for two 688 bearings each (end idlers, M2). Rev B sheet 21.

Link: MISUMI SG: HTD 5M timing pulleys

https://sg.misumi-ec.com/vona2/detail/110310532889/

# Side plate, front segment (Rev B P1)



Quantity:  2 (1 + 1 opposite hand)

**MAKE: laser-cut aluminium (Nhien's)**

## What it does

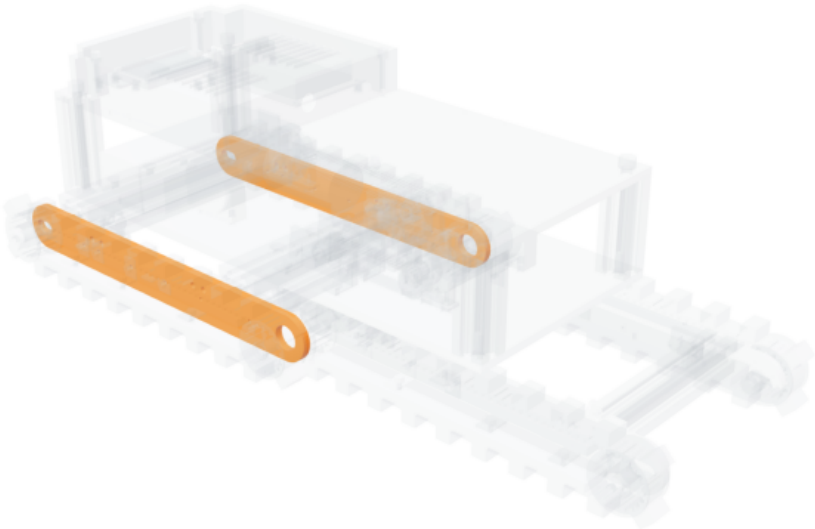
255 x 30.4 plate, inboard (z 55-60). Ø16 hole for the hinge bearing, a slot at the front end for the idler's shoulder screw (slide it to tension the track), tapped holes for the rail brackets and the 2020 bar brackets.

## Where to get it

5 mm Al 6061, laser cut from DXF/P1 Side plate - FRONT segment.dxf, then drill and tap per Rev B sheet 4. Fab Lab laser does not cut aluminium: Professional Workshop or an outside laser shop on the vendor list.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

# Side plate, rear segment (Rev B P2)



Quantity:  2 (1 + 1 opposite hand)

**MAKE: laser-cut aluminium (Nhien's)**

## What it does

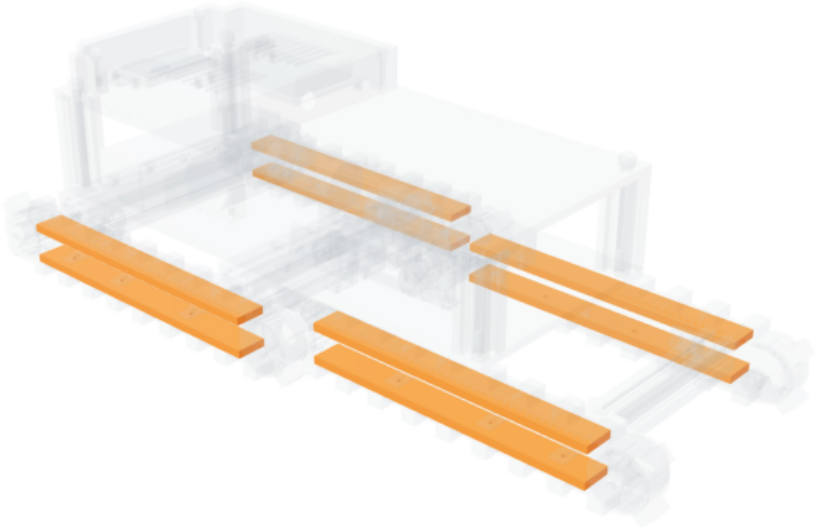
Outboard plate (z 80-85). The rear track runs outside it on a shoulder screw. Holes for the 3030 bar brackets and the rails.

## Where to get it

As P1, from DXF/P2 Side plate - REAR segment.dxf, Rev B sheet 5.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

# Slide rails (Rev B P3-P6)



Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Quantity: 8

**BUY strip + drill**

## What it does

The strips the track's upper and lower runs slide on so the track cannot sag between the wheels. Does the job my printed rail did. My track's tooth tips ride on the lower one.

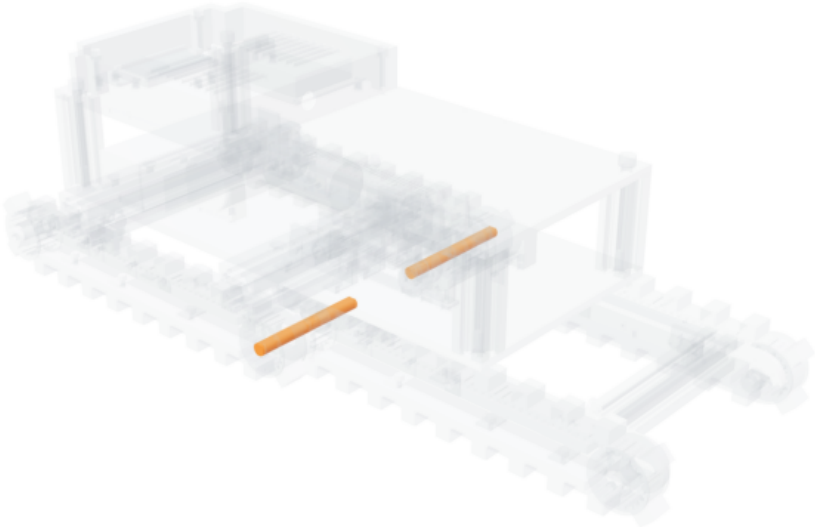
## Where to get it

MISUMI SG UHMW-PE strip UPA-189-20-5 (link), 189 x 20 x 5, 3 pilot holes each, Rev B sheet 18. Held to the plates by RS PRO 427-991 angle brackets (RS SG) with M3 screws.

Link: MISUMI SG: plastic strips (UHMW)

https://sg.misumi-ec.com/vona2/mech/M1500000000/M1501000000/

# Hinge shaft (Rev B P20 A + B)



Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Quantity:  2 (or 1 if one motor)

**BUY D-shaft, cut**

## What it does

Replaces my 230 mm middle axle. The drive pulleys sit on the D-flat, so no set screws. Runs in 688 bearings in both plates and in the P17 support.

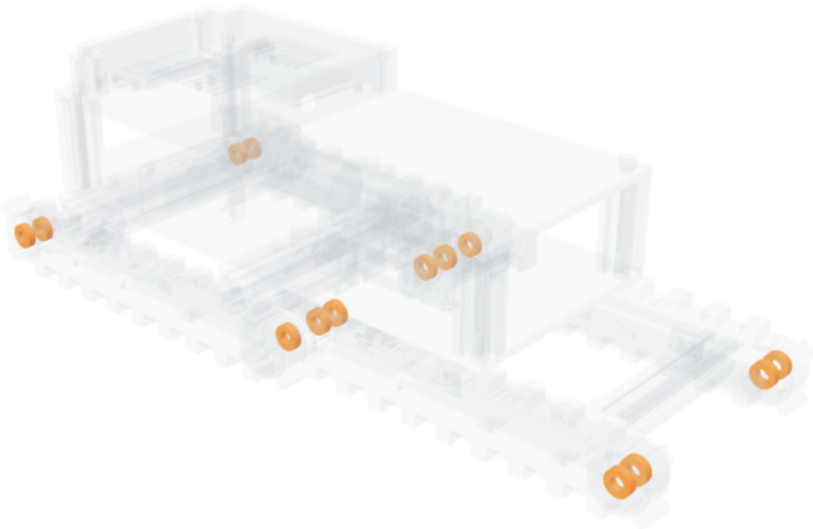
## Where to get it

Ø8 h7 D-shaft, flat 7, 200 mm stock (MISUMI SG or RS SG linear shafts, link). Cut to 82.3 (A) and 77.3 (B); E-clip grooves per Rev B sheet 20. One motor per side -> two half shafts; one motor -> one 212 mm shaft.

Link:  RS SG: linear shafts

https://sg.rs-online.com/web/c/bearings-seals/linear-motion/linear-shafts/

# Ball bearing 688-2RS 8 x 16 x 5



Quantity: 14

**BUY: RS Singapore**

## What it does

Replaces my 10 x 608ZZ. Smaller OD (16) because the plate is only 30 mm tall.

## Where to get it

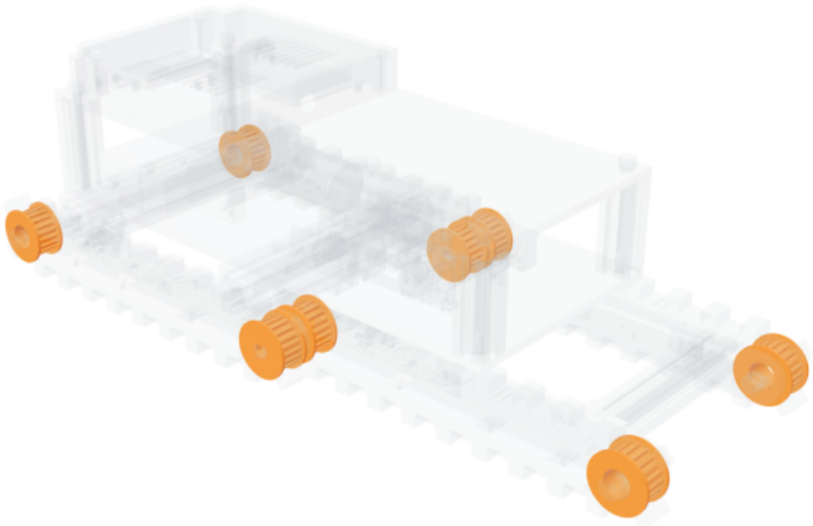
RS SG ball bearings (link): search 688-2RS. 4 in the plate hinge holes, 2 in the P17 supports, 8 inside the four idler pulleys.

Link: RS SG: ball bearings

https://sg.rs-online.com/web/c/bearings-seals/bearings/ball-bearings/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

# Idler stub axle: shoulder screw Ø8 x 25, M6



Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Quantity: 4

**BUY: MISUMI SG or RS SG**

## What it does

Replaces my front and rear axles. One per plate end, through the slot; the end wheel spins on two 688s on its shoulder. Slide outwards in the slot and tighten the nut to tension the track.

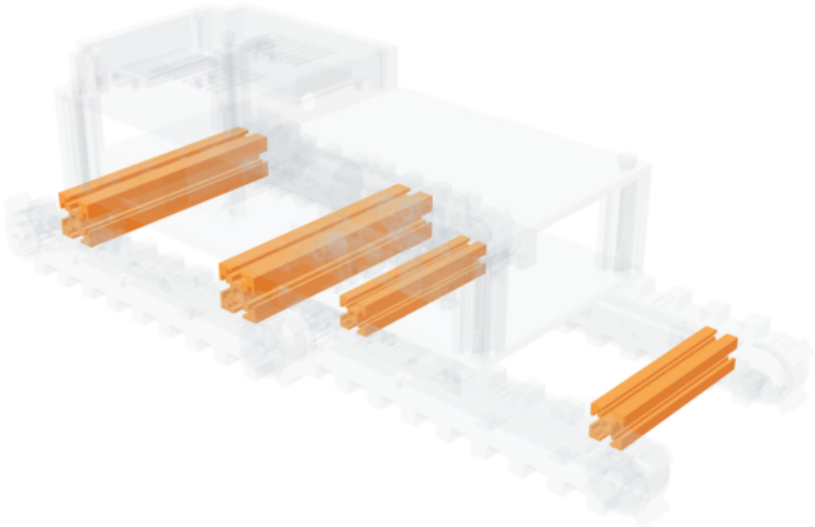
## Where to get it

ISO 7379 8 x 25 shoulder screw (link), with 8 x 11 x 1 washer, 8 x 11 x 0.5 shim, 8 x 10 x 9 spacer, M6 washer and nyloc. Rev B sheet 3, idlers.

Link: RS SG: shoulder screws

https://sg.rs-online.com/web/c/fasteners-fixings/screws-bolts/shoulder-screws/

# Cross bars: 2020 x 110 and 3030 x 160 (Rev B P7, P8)



Quantity:  2 + 2

**BUY profile, cut**

## What it does

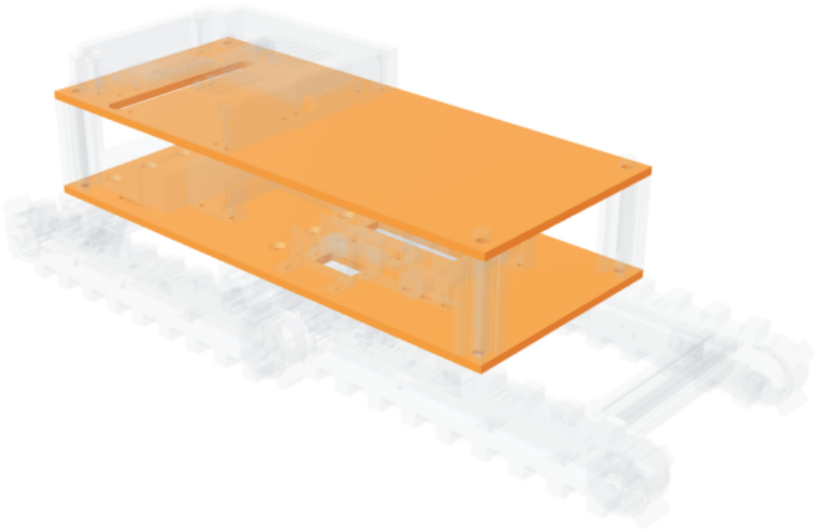
Replace my M4 rods and printed spacers: they hold the plates apart and carry the deck. The 3030 bars are also what the box/deck bolts to.

## Where to get it

2020 B-type slot 6 (1 m) and 3030 slot 8 (0.5 m) from MISUMI SG or a local supplier; cut per Rev B sheet 19. Corner brackets P18/P19 from aluminium angle, T-nuts M5/M6.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

# Deck and tray (Rev B P12, P16) and everything on them



Quantity: 1 + 1

**Nhien's parts**

## What it does

The 'box': 160 mm wide between the rear plates in this layout. Motors, drivers, battery and the platform mount sit here.

## Where to get it

Al 5 mm laser cut, Rev B sheets 6-8; posts P15, controller cover P22, motors, electronics per the Rev B BOM.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

# Procurement, hybrid build: my track on Nhien's Rev B frame

Date 15 Oct 2026. Prices S$ ex GST as seen in listings; confirm at checkout. Rev B = Nhien's pack (`RB4107 Rev B`), whose BOM covers everything in section 2.

## 1. Mine (track side)

| Part | Qty | Make / buy | Where | Price | Arrives |
|---|---|---|---|---|---|
| Track, TPU 95A, `stl/track_TPU_x4.stl` | 4 | **Make**: Fab Lab UP300 (TPU head) | Element14 SG TPU 95A filament if the lab has none | ~S$45/kg, uses 130 g | filament 1-2 days; 3 h print each |
| Fit-test slice, `stl/fit_test_track_5mm_TPU.stl` | 1 | Make | same | - | 10 min |
| Stand-in pulley, `stl/stand_in_pulley_20T_PLA_test_only.stl` | 1 | Make, PLA | any printer | - | 1 h |
| Timing pulley HTD 5M 20T, 15 mm belt, flanged, **plain (pilot) bore** | 8 | **Buy** | MISUMI SG, Timing Pulleys HTD 5M | ~S$20 each | ships 1 day from SG stock; allow 2-5 days |
| Pulley bore modification: 4 x D-bore Ø8 (M1), 4 x Ø16 H7 (M2) | 8 | **Make**: Professional Workshop request, Rev B sheet 21 | E3-01-06 | free | workshop queue, allow 1-2 weeks |

## 2. Nhien's (frame, drive, hinge, electronics), from the Rev B BOM

| Part | Qty | Make / buy | Where |
|---|---|---|---|
| Side plates P1 (front) and P2 (rear), 5 mm Al, laser + tap | 2 + 2 | Make | DXF in the pack; Professional Workshop or a laser shop on the vendor list |
| Slide rails P3-P6, UHMW strip 189 x 20 x 5 | 8 | Buy + drill | MISUMI SG UPA-189-20-5 |
| Rail brackets RS PRO 427-991 (angle 15 x 14.6 x 10) | 32 | Buy | RS SG stock no. 427-991 |
| Hinge shafts P20 A 82.3 + B 77.3 (or one 212), Ø8 D-shaft flat 7 | 2 | Buy stock, cut | RS SG linear shafts or MISUMI SG |
| Bearing 688-2RS 8 x 16 x 5 | 14 | Buy | RS SG ball bearings |
| Shoulder screw ISO 7379 Ø8 x 25 M6 + washers, shim, spacer, nyloc | 4 sets | Buy | RS SG shoulder screws |
| 2020 profile x 110 (P7) x2, 3030 x 160 (P8) x2, corner posts P15 x4 | 1 m + 0.5 m | Buy, cut | MISUMI SG / local |
| Corner brackets P18 (25x25x4 angle) x4, P19 (20x20x3) x4, T-nuts, E-clips, shims | - | Buy / make | Rev B BOM sections 3-4 |

| | | | |
|---|---|---|---|
| Deck P12, tray P16, cover P22, motors, drivers, ESP32, battery | - | Theirs | Rev B BOM |

## 3. What is no longer needed from my earlier list

Printed side plates (4), printed spacers (6), M4 rods and nuts, Ø8 plain rod for three axles, 608ZZ x10, M4 set screws. Keep the STL files in case the team falls back to the all-printed frame.

## 4. Order of events

1. Order the 8 plain-bore pulleys today (critical path, then the workshop bore mod); print the stand-in pulley and fit-test slice; print the 4 tracks as soon as a UP300 slot is free.
2. Nhien: laser plates, order rails, bearings, shoulder screws, profiles.
3. Workshop: pulley bores, plate tapping, shaft cutting (one request form).
4. Assemble per Rev B sheet 3; fit my tracks on the real pulleys; tension at the slots.

# Layout with Nhien's side plates (Rev B P1 / P2): what changes on the track side

Read from `RB4107 Rev B/STEP` (plates P1, P2 and the full assembly, 15 Oct 2026). Rev B's frame origin is the hinge axis with the A side at z < 60; below everything is converted to **z from the vehicle centre line**, one side (the other mirrors), mm at 1:5.

## Their plates

| | Rev B plate | My printed plate (for comparison) |
|---|---|---|
| Material | 5 mm aluminium 6061, laser cut, holes tapped | 7 mm PETG, printed |
| Size | 255 x 30.4 (R15 ends) | 265 x 45 (R20 ends) |
| Axle spacing | 225 (hinge hole to the middle of the idler slot) | 225 |
| Hinge hole | Ø16 for a 688-2RS bearing (8 x 16 x 5) | Ø22.2 for a 608ZZ |
| End hole | slot 220.4-232.6 for an Ø8 shoulder-screw stub axle: slide to tension the track | round pocket |
| Rail | separate UHMW strips P3-P6, screwed on through the C holes with small brackets | printed as part of the plate |
| Cross bars | 2020 profile x 110 (front), 3030 profile x 160 (rear), corner brackets + T-nuts | M4 rod + printed spacer |

## Where everything sits across the width (per side, from the centre line)

| z | What |
|---|---|
| 0 to 24 (A side) / 29 (B side) | free: hinge shaft ends here, hinge pinion / sector gear region |
| 36 to 41 | P17 hinge-shaft bearing support (688 bearing), bolted to the middle 3030 bar |
| **55 to 60** | **front plate P1** (688 bearing in it at the hinge) |
| 60.5 to 79.5 | front segment track lane: pulley 19 over flanges, track 62.5 to 77.5 |
| **80 to 85** | **rear plate P2** (688 bearing at the hinge) |
| 85.5 to 104.5 | rear segment track lane, **outboard of the rear plate**; track 87.5 to 102.5 |
| 106 | end of the hinge shaft |

So: 205 over the track faces, 209 over the pulley flanges, and the frame between the rear plates is 160 wide inside (not 80 as in my layout): the box can be up to 160 wide.

## Consequences for the parts I was making

| Part | With their plates |
|---|---|

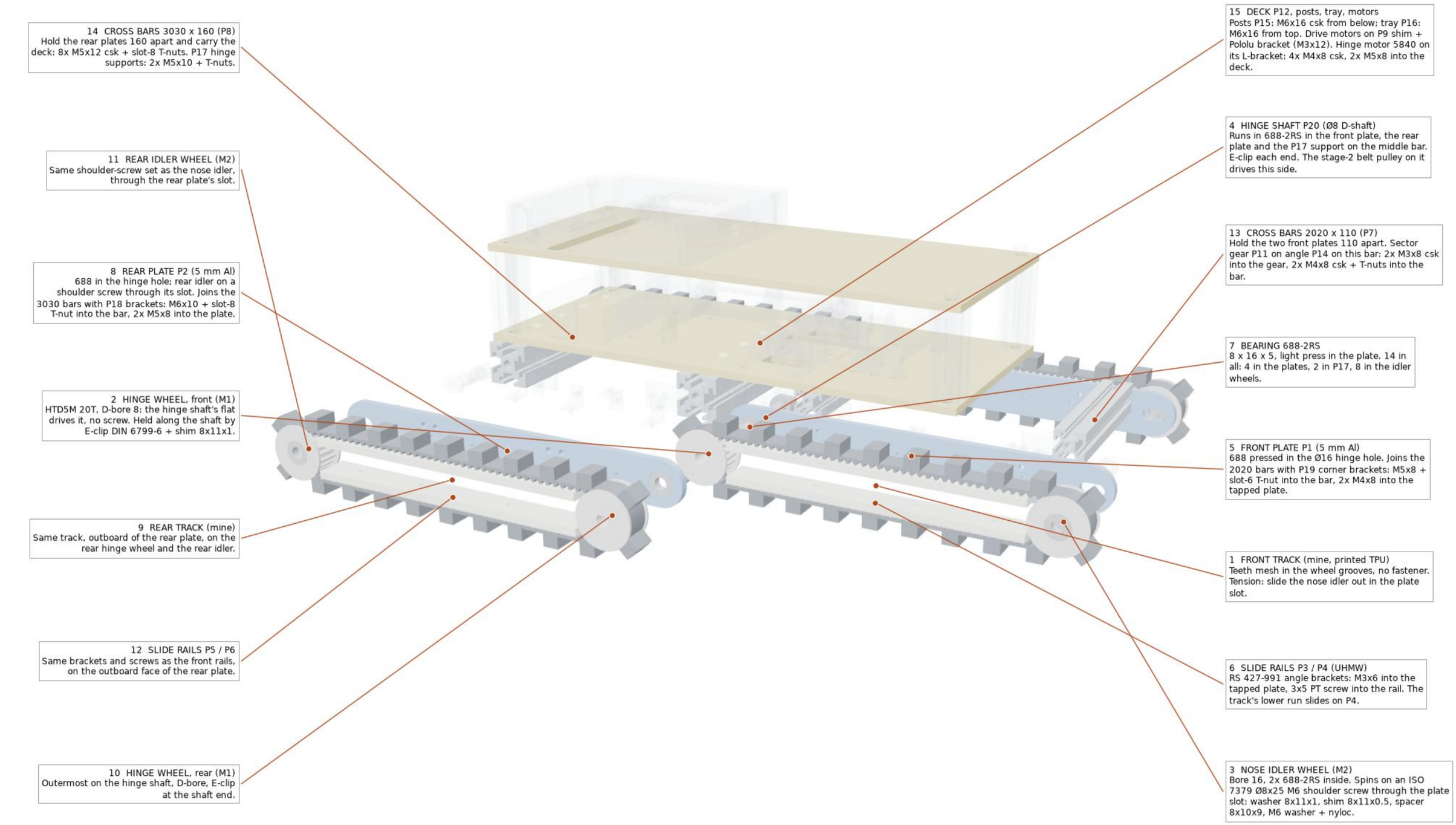
| | |
|---|---|
| **Track x4** | unchanged: 550 loop, 110 teeth, 15 wide, 45 cleats. Still printed. The only printed part left. |
| **Side plates** | theirs (P1, P2). Not printed. |
| **Spacers** | **none.** The 2020 and 3030 bars hold the plates apart. |
| **Cross bar rods, nuts, washers** | **none.** Replaced by the bars, P18/P19 corner brackets, T-nuts. |
| **Middle axle 230** | replaced by Rev B's two hinge half-shafts P20: Ø8 D-shaft (flat 7), A = 82.3 long (z 14 to 106 on the A side), B = 77.3 (z 31 to 106 on the B side). Two halves because Rev B has one motor per side. If the team goes to **one motor**, use one Ø8 D-shaft 212 long (z -106 to +106) instead, same stock. |
| **Front and rear axles** | **none.** The end pulleys ride on Ø8 x 25 M6 shoulder screws (ISO 7379) through the plate slots, one per plate, 4 in all, each with two 688-2RS inside a Ø16-bored pulley. |
| **Bearings** | 608ZZ x10 -> **688-2RS x14** (8 x 16 x 5): 4 in the plate hinge holes, 2 in the P17 supports, 8 in the idler pulleys. |
| **Pulleys** | same HTD 5M 20T 15 mm belt, but the lanes are only 19-20 wide, so there is **no room for a set-screw hub**. Rev B modifies them: the 4 on the hinge shaft get a D-bore 8 (M1), the 4 idlers get a Ø16 H7 bore for the 688s (M2). That is lathe work at the Professional Workshop; order plain-bore (pilot-bore) pulleys for it. |
| **Set screws** | none on the pulleys (D-bore drives them); E-clips DIN 6799-6 and a shim on each shaft end instead. |

## What I need from Nhien to finish the hybrid

1. One motor or two? It decides whether the hinge shaft is one piece (212) or two (82.3 + 77.3).
2. Confirm the rail strips P3-P6 are coming (MISUMI UPA-189-20-5 UHMW, 8 off) or whether the lower run just rides on nothing; my track's tooth tips sit 13.34 below the axle and need the rail.
3. The track is still mine to print; the step-matched cleat pitch (12.22) replaces Rev B's 22 cleats at 25 mm on a stock belt. Both fit the same pulleys, so either can be tested on their frame.

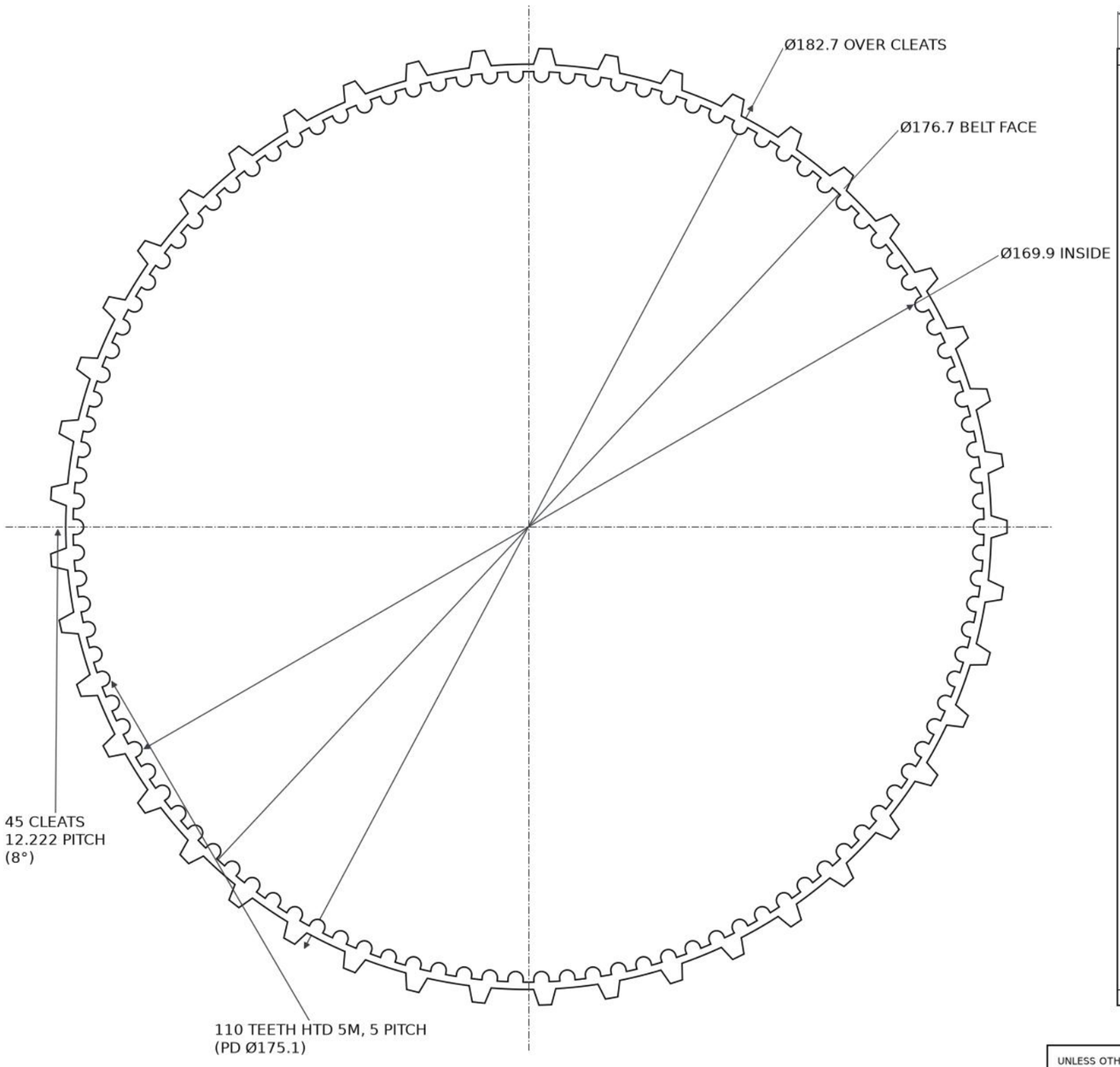
# How it all connects: one side pulled apart, every joint and its fastener

Nhien's Rev B frame with my printed track. Near side exploded outwards from the hinge shaft (plate, then track and wheels, then rear plate, then rear track); far side left assembled. Bought parts grey, Nhien's blue, deck yellow, electronics ghosted.




Assembly order (Rev B sheet 3): 1 bearings into plates, P17 and idlers  2 rear frame: P2 + 3030 bars + P18  3 front frame: P1 + 2020 bars + P19 + sector gear  4 hinge shaft through rear wheel, rear plate, front wheel, front plate, P17; E-clips  5 deck, posts, tray  6 motors, belts, tension  7 idlers on shoulder screws, tension the tracks  8 electronics, cover.  Fasteners from Rev B sheet 3; my track and its fit test from this pack.

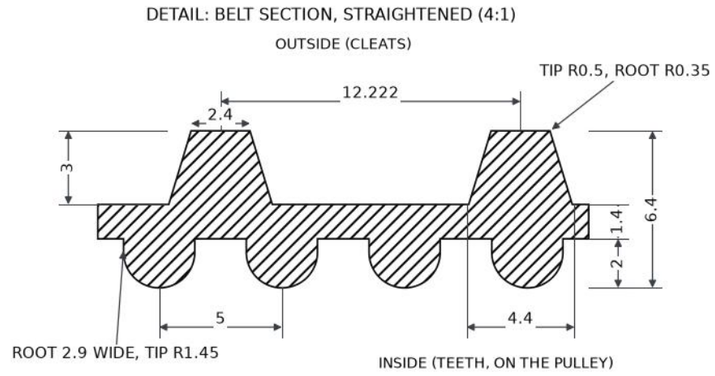
FRONT VIEW




EDGE VIEW




DETAIL: BELT SECTION, STRAIGHTENED (4:1)




NOTES:
1. FILE stl/track_TPU_x4.stl. PRINT STANDING AS A RING, NO SUPPORTS, 0.2 LAYERS, 100% INFILL, 20-30 MM/S, DIRECT-DRIVE EXTRUDER.
2. LOOP 550 ON THE PITCH LINE = 110 TEETH x 5 PITCH; RUNS ON TWO HTD 5M 20T PULLEYS (TS-06) AT 225 CENTRES. ABOUT 32 G.
3. 45 CLEATS AT 12.222 PITCH: 5 CLEATS = 61.11, ONE ES1 STEP ALONG THE SLOPE (35 x 50 AT 1:5), SO EVERY STEP EDGE SITS BETWEEN TWO CLEATS.
4. BED SMALLER THAN 190 MM: ASK FOR THE TWO-PIECE TRACK. PRINT THE 5 MM FIT-TEST SLICE FIRST (stl/fit_test_track_5mm_TPU.stl).

THIRD ANGLE



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**TRACK, 110T HTD 5M, 45 CLEATS**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: TPU 95A, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-01

QTY.: 4

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

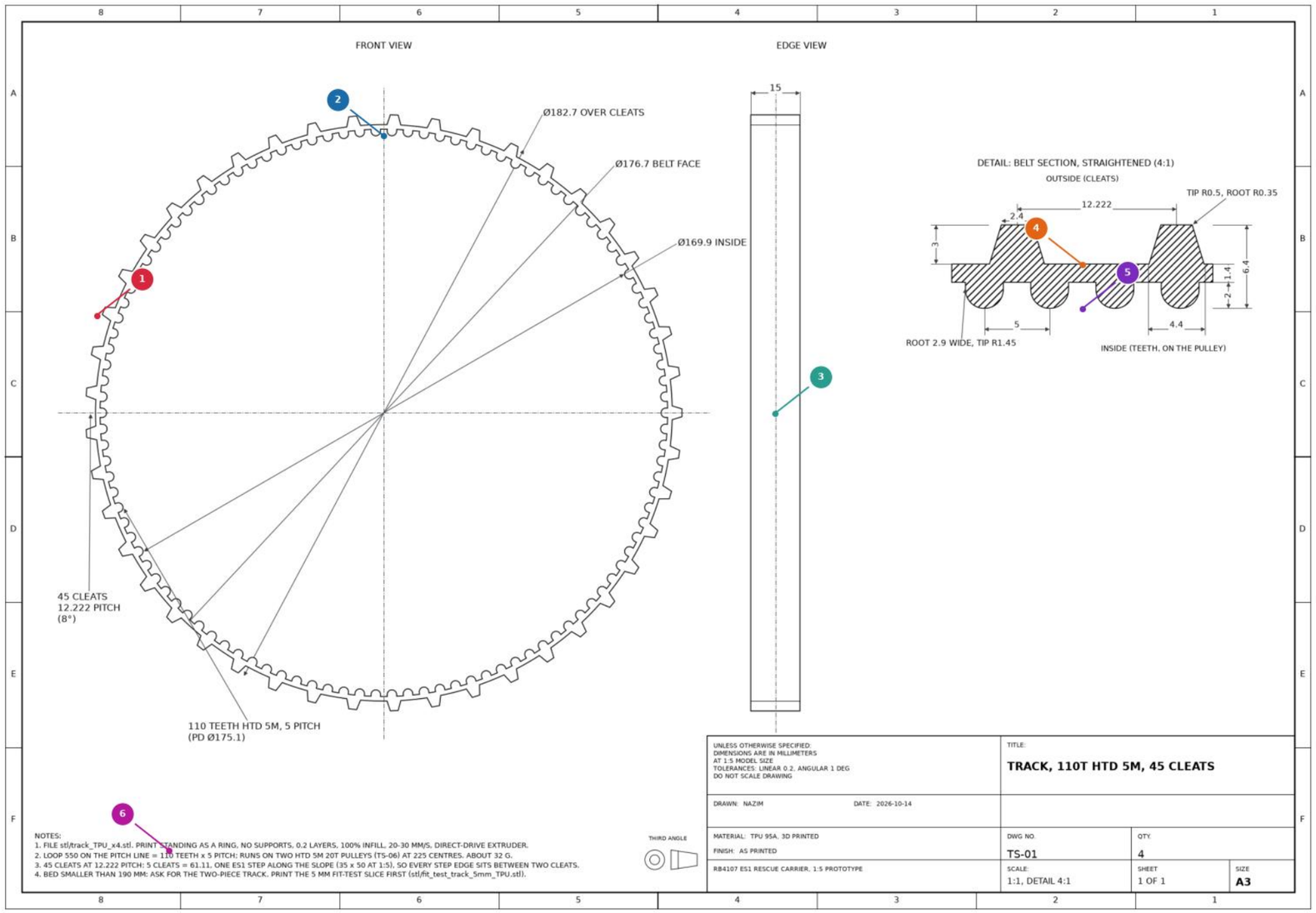
SCALE: 1:1, DETAIL 4:1

SHEET: 1 OF 1

SIZE: **A3**

# How to read TS-01: the track

Markers below sit on a reduced copy of sheet TS-01 (previous page, full size). Numbers match the key on the right.




**1 Cleats, outside**

45 bumps 12.22 apart (8 degrees). Five of them span one ES1 step, so every step edge lands between two cleats and the track holds like a gear on a rack.

**2 Teeth, inside**

110 HTD 5M teeth, 5 apart. These mesh with the 20-tooth wheels; this is what stops the track slipping.

**3 Edge view**

The ring seen from the side: 15 wide, 182.7 tall over the cleats.

**4 Cleat profile (4:1)**

A cleat is 3 tall, 4.4 wide at its base and 2.4 at the tip, with rounded corners so it releases from the step.

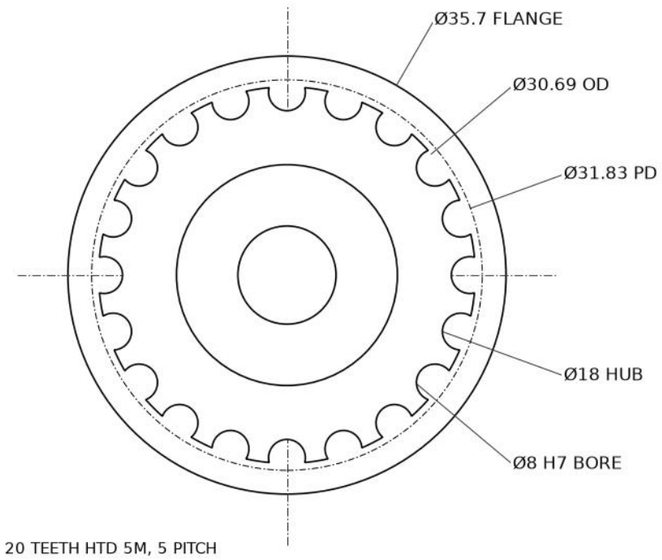
**5 Tooth profile (4:1)**

A tooth is 2 deep with a 1.45 radius tip and a 2.9 root; the 1.4 backing between teeth and cleats is the belt itself. Whole section 6.4 thick.

**6 Notes**

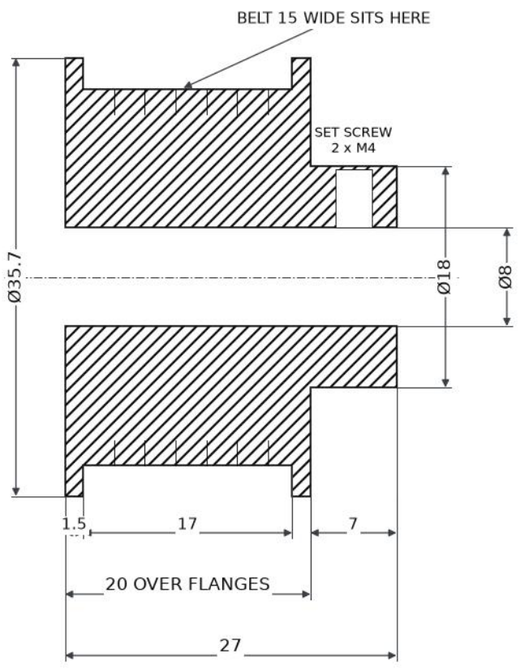
Print standing up as a ring, no supports, 100% infill, TPU 95A, on the Fab Lab UP300. Four tracks, about 32 g each.

FRONT VIEW




20 TEETH HTD 5M, 5 PITCH

SECTION (ALONG THE AXLE)




NOTES:
1. BOUGHT PART. ORDER: HTD 5M, 20 TEETH, FOR 15 MM BELT, 8 MM BORE, FLANGES BOTH SIDES, SET-SCREW HUB. MISUMI SINGAPORE (APPROVED VENDOR). 4 PER SEGMENT.
2. WHAT MATTERS: 20T HTD 5M (PD 31.83), 15 MM BELT, 8 BORE. FLANGE, HUB AND LENGTH SHOWN ARE THE STAND-IN'S: CHECK THE VENDOR DRAWING.
3. LAYOUT ALLOWS 20 OVER THE FLANGES AND 7 FOR THE HUB (27 OVERALL) PER PULLEY. A LONGER PULLEY MOVES THE LANES: TELL THE TEAM BEFORE ORDERING.
4. PLA STAND-IN FOR FIT TESTS ONLY: stl/stand_in_pulley_20T_PLA_test_only.stl (M3 SET SCREW). NOT FOR DRIVING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**TIMING PULLEY HTD 5M 20T (BOUGHT)**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: ALUMINIUM (VENDOR'S)
FINISH: VENDOR'S

DWG NO.: TS-06

QTY.: 8

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

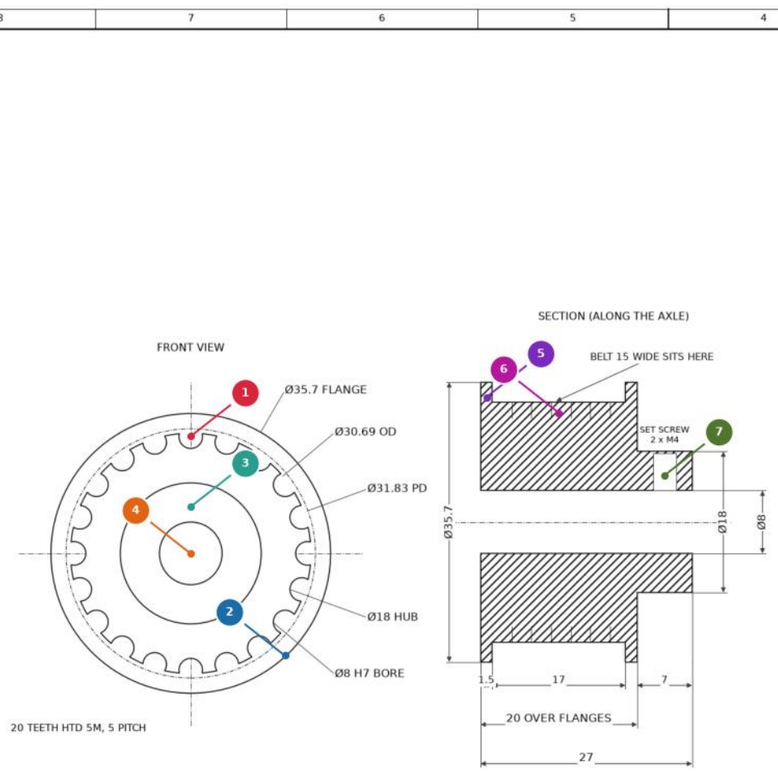
SCALE: 2:1

SHEET: 1 OF 1

SIZE: **A3**

# How to read TS-06: the toothed wheel (bought)

Markers below sit on a reduced copy of sheet TS-06 (previous page, full size). Numbers match the key on the right.




**1 — 20 grooves**

The track's teeth drop into these. HTD 5M profile, 5 pitch, pitch diameter 31.83.

**2 — Flange**

A thin Ø35.7 disc on each side of the teeth. Keeps the 15 wide track from walking off sideways.

**3 — Hub**

Ø18 boss, 7 long, on one side. The 2 M4 set screws sit in it and clamp the wheel to the axle.

**4 — Bore**

Ø8 H7: a close sliding fit on the Ø8 axle.

**5 — Flange, in section**

1.5 thick, both sides. Total 20 over the flanges, which is the lane width in the layout.

**6 — Belt seat**

17 between the flanges; the 15 wide track sits here with 1 each side.

**7 — Hub and set screws**

The 7 long hub with 2 x M4 set screws onto the axle flat. 27 overall length per wheel.

NOTES:
1. BOUGHT PART. ORDER: HTD 5M, 20 TEETH, FOR 15 MM BELT, 8 MM BORE, FLANGES BOTH SIDES, SET-SCREW HUB. MISUMI SINGAPORE (APPROVED VENDOR). 4 PER SEGMENT.
2. WHAT MATTERS: 20T HTD 5M (PD 31.83), 15 MM BELT, 8 BORE, FLANGE, HUB AND LENGTH SHOWN ARE THE STAND-IN'S: CHECK THE VENDOR DRAWING.
3. LAYOUT ALLOWS 20 OVER THE FLANGES AND 7 FOR THE HUB (27 OVERALL) PER PULLEY. A LONGER PULLEY MOVES THE LANES: TELL THE TEAM BEFORE ORDERING.
4. PLA STAND-IN FOR FIT TESTS ONLY: stl/stand_in_pulley_20T_PLA_test_only.stl (M3 SET SCREW). NOT FOR DRIVING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: ALUMINIUM (VENDOR'S)
FINISH: VENDOR'S

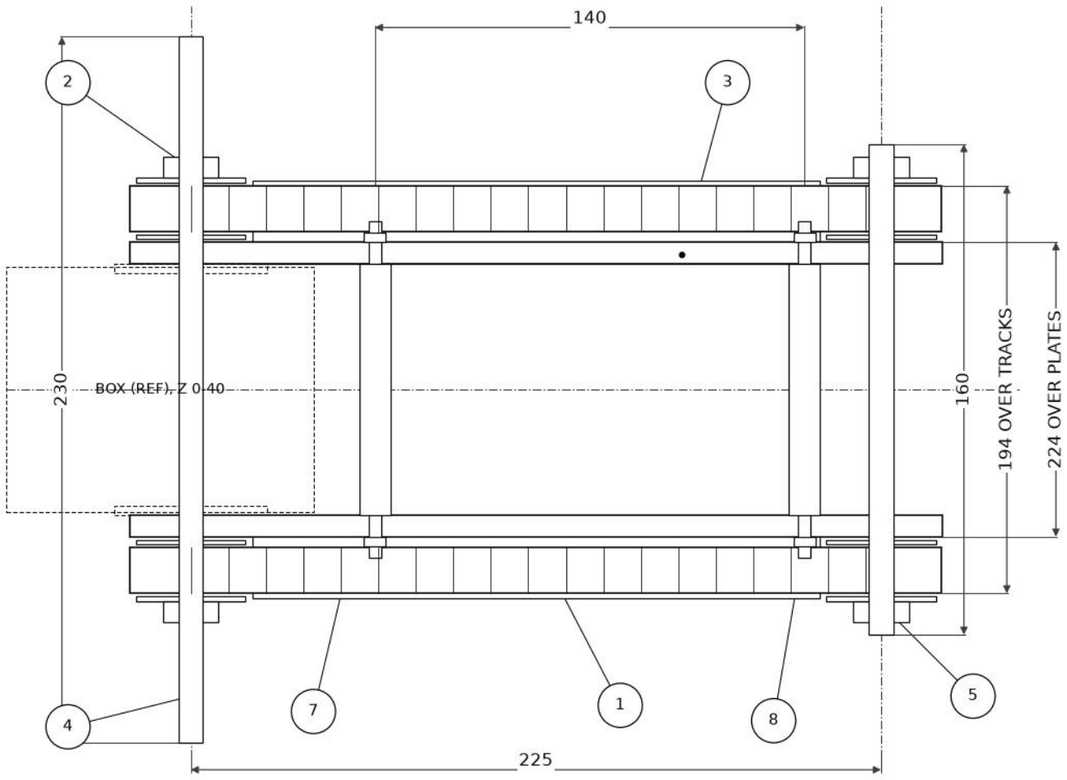
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

TITLE: **TIMING PULLEY HTD 5M 20T (BOUGHT)**

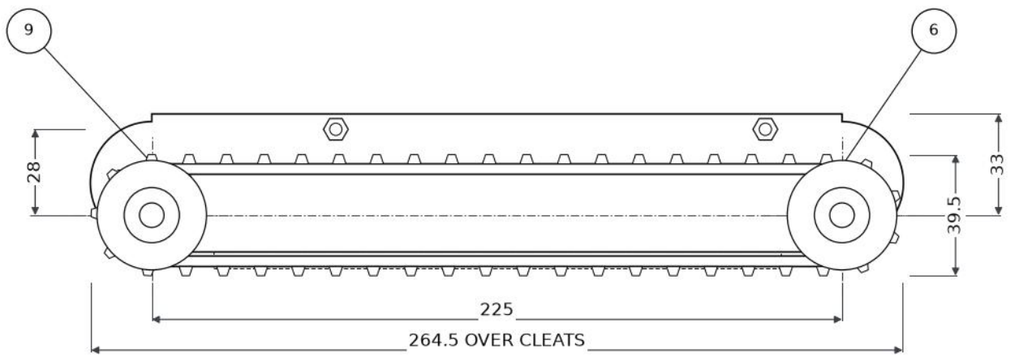
DWG NO. TS-06
QTY. 8
SCALE: 2:1
SHEET 1 OF 1
SIZE **A3**

| ITEM NO. | PART NUMBER | DESCRIPTION | QTY. |
|---|---|---|---|
| 1 | TS-01 TRACK | TRACK, TPU 95A, 110T HTD 5M, 45 CLEATS | 2 |
| 2 | TS-06 PULLEY_20T | PULLEY HTD 5M 20T, 15 MM BELT, BORE 8, BOUGHT | 4 |
| 3 | TS-02 SIDE_PLATE_FRONT | SIDE PLATE, FRONT SEGMENT, PETG, 1 L + 1 R | 2 |
| 4 | TS-04 AXLE_MIDDLE | MIDDLE AXLE, ROD 8 x 230, SHARED W/ REAR SEG | 1 |
| 5 | TS-04 AXLE_FRONT | FRONT AXLE, ROD 8 x 160 | 1 |
| 6 | TS-07 BEARING_608ZZ | BALL BEARING 608ZZ, 8 x 22 x 7, BOUGHT | 4 |
| 7 | TS-05 SPACER_82 | SPACER OD 10 x 82, BORE 4.5, PETG | 2 |
| 8 | M4_ROD_110 | M4 THREADED ROD x 110 + 2 NUTS + 2 WASHERS | 2 |
| 9 | M3x14 | M3 x 14 BOLT + NUT, HINGE SECTOR GEAR, BOUGHT | 6 |

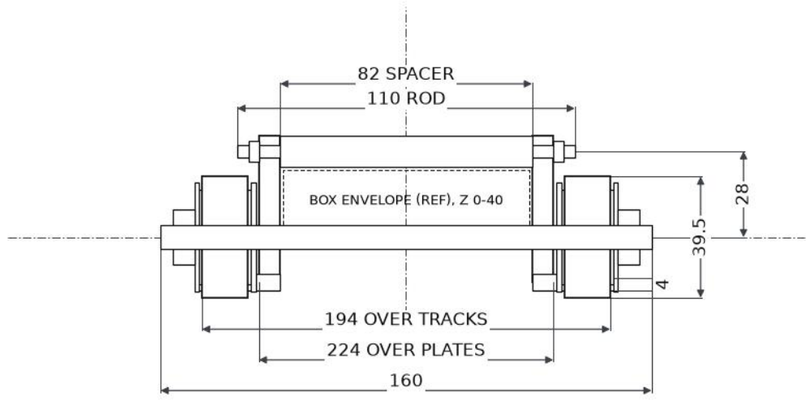
TOP VIEW




SIDE VIEW (FROM OUTBOARD)




END VIEW (FROM THE FRONT)




NOTES:
1. REAR SEGMENT: SAME, WITH ITS PLATES OUTBOARD (Z 105-112, TS-03), RAILS INBOARD, AND THE REAR AXLE (TS-04) INSTEAD OF THE FRONT AXLE.
2. BOTH SEGMENTS SHARE THE MIDDLE AXLE (ITEM 4), TURNED BY THE DRIVE MOTOR. PULLEYS ARE SET-SCREWED TO THE AXLES; THE AXLES TURN IN ITEM 6.
3. HINGE SECTOR GEAR (3 THICK, Z 38-41, NOT SHOWN) BOLTS TO THE INBOARD FACE OF EACH FRONT PLATE WITH ITEM 9. THE BOX STOPS AT Z 37 THERE.
4. THE BOX (TEAM'S) MUST END BEFORE X 50 SO THE FRONT CROSS BAR AT X 60 CLEARS IT WHEN THE HINGE BENDS. MM AT 1:5; X 5 FOR FULL SIZE.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**FRONT TRACK SEGMENT ASSEMBLY**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: SEE PARTS LIST
FINISH:

DWG NO.
TS-00

QTY.
1 (REAR SIMILAR)

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

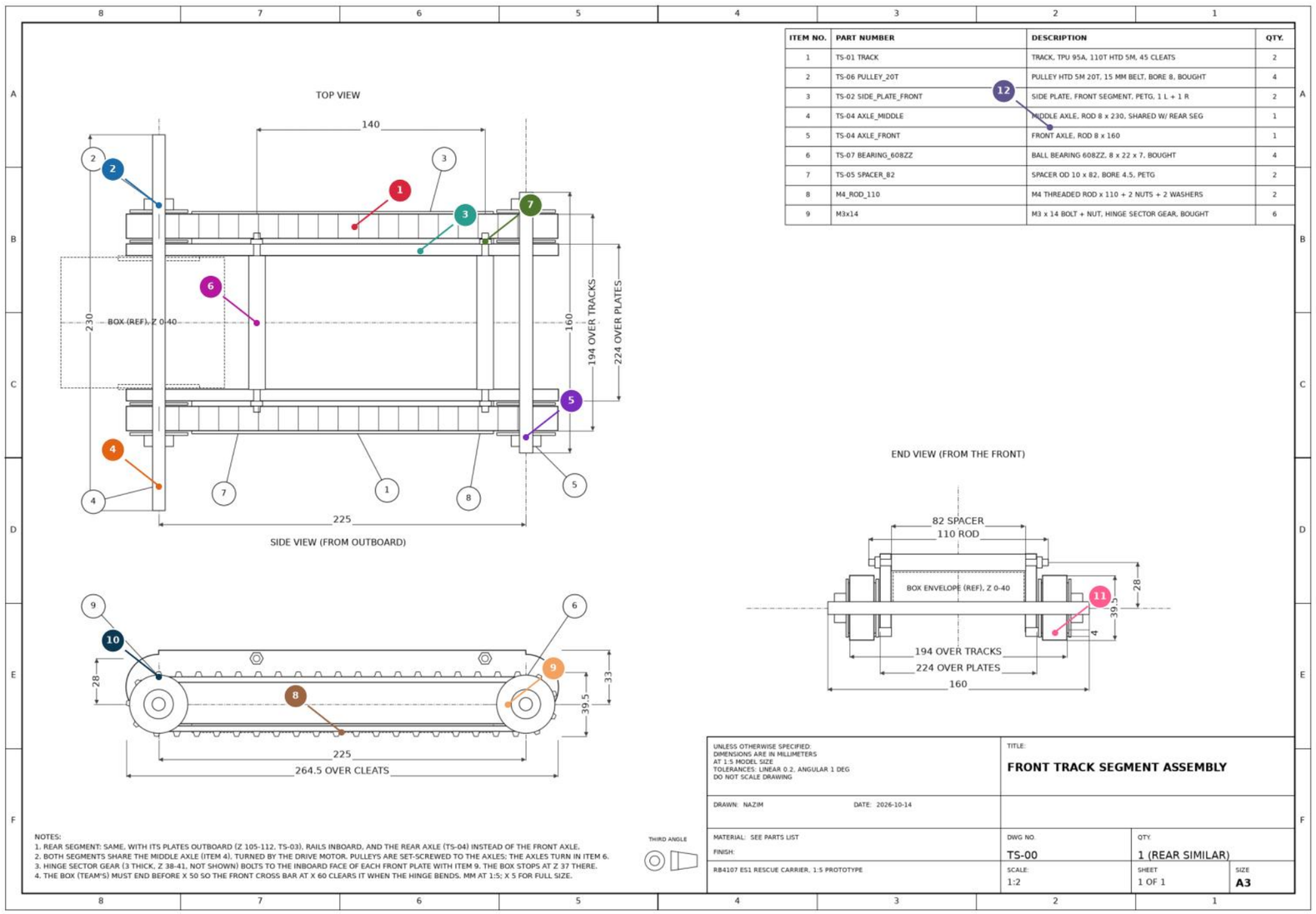
SCALE:
1:2

SHEET
1 OF 1

SIZE
**A3**

# How to read TS-00: the front track segment put together

Markers below sit on a reduced copy of sheet TS-00 (previous page, full size). Numbers match the key on the right.




| ITEM NO. | PART NUMBER | DESCRIPTION | QTY. |
|---|---|---|---|
| 1 | TS-01 TRACK | TRACK, TPU 95A, 110T HTD 5M, 45 CLEATS | 2 |
| 2 | TS-06 PULLEY_20T | PULLEY HTD 5M 20T, 15 MM BELT, BORE 8, BOUGHT | 4 |
| 3 | TS-02 SIDE_PLATE_FRONT | SIDE PLATE, FRONT SEGMENT, PETG, 1 L + 1 R | 2 |
| 4 | TS-04 AXLE_MIDDLE | MIDDLE AXLE, ROD 8 x 230, SHARED W/ REAR SEG | 1 |
| 5 | TS-04 AXLE_FRONT | FRONT AXLE, ROD 8 x 160 | 1 |
| 6 | TS-07 BEARING_608ZZ | BALL BEARING 608ZZ, 8 x 22 x 7, BOUGHT | 4 |
| 7 | TS-05 SPACER_82 | SPACER OD 10 x 82, BORE 4.5, PETG | 2 |
| 8 | M4_ROD_110 | M4 THREADED ROD x 110 + 2 NUTS + 2 WASHERS | 2 |
| 9 | M3x14 | M3 x 14 BOLT + NUT, HINGE SECTOR GEAR, BOUGHT | 6 |

NOTES:
1. REAR SEGMENT: SAME, WITH ITS PLATES OUTBOARD (Z 105-112, TS-03), RAILS INBOARD, AND THE REAR AXLE (TS-04) INSTEAD OF THE FRONT AXLE.
2. BOTH SEGMENTS SHARE THE MIDDLE AXLE (ITEM 4), TURNED BY THE DRIVE MOTOR. PULLEYS ARE SET-SCREWED TO THE AXLES; THE AXLES TURN IN ITEM 6.
3. HINGE SECTOR GEAR (3 THICK, Z 38-41, NOT SHOWN) BOLTS TO THE INBOARD FACE OF EACH FRONT PLATE WITH ITEM 9. THE BOX STOPS AT Z 37 THERE.
4. THE BOX (TEAM'S) MUST END BEFORE X 50 SO THE FRONT CROSS BAR AT X 60 CLEARS IT WHEN THE HINGE BENDS. MM AT 1:5; X 5 FOR FULL SIZE.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE: **FRONT TRACK SEGMENT ASSEMBLY**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: SEE PARTS LIST
FINISH:

DWG NO. TS-00
QTY. 1 (REAR SIMILAR)

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 1:2
SHEET 1 OF 1
SIZE A3

## 1 Track (item 1)

The printed TPU belt. Its inside teeth sit on the two toothed wheels; its outside cleats grip the steps. One each side.

## 2 Pulley = toothed wheel (item 2)

Bought HTD 5M 20-tooth wheel. Sits on the axle and is locked to it by 2 M4 set screws into a filed flat. Four per segment.

## 3 Side plate (item 3)

Printed PETG plate, one each side. Holds the two bearings the axles turn in and carries the slide rail under the track.

## 4 Middle axle (item 4)

Ø8 steel rod, 230 long, shared with the rear segment. It is the hinge: both segments' plates turn on it. The motor drives it.

## 5 Front axle (item 5)

Ø8 rod, 160 long. Only this segment's two front pulleys sit on it. It just idles.

## 6 Cross bar spacer (item 7)

Printed Ø10 tube, 82 long, between the two plates on each M4 rod. Sets the plate spacing.

## 7 M4 rod, nut and washer (item 8)

M4 threaded rod 110 long through plate, spacer, plate; a washer and nut on each end pull the plates onto the spacer. Two per segment.

## 8 Track loop, side view

The belt wraps the two wheels; cleats stand out of the loop. The flat lower run rests on the plate's rail (dashed, hidden inside the loop).

## 9 Bearing 608ZZ (item 6)

Pressed into the plate's Ø22.2 pocket, dashed because it is hidden behind the pulley flange. The axle runs through its 8 mm bore.

## 10 Sector gear bolts (item 9)

Three Ø3.4 holes on a 17 mm circle round the middle axle. The hinge sector gear (team's part) bolts to the plate's inboard face with M3 x 14 and nuts.

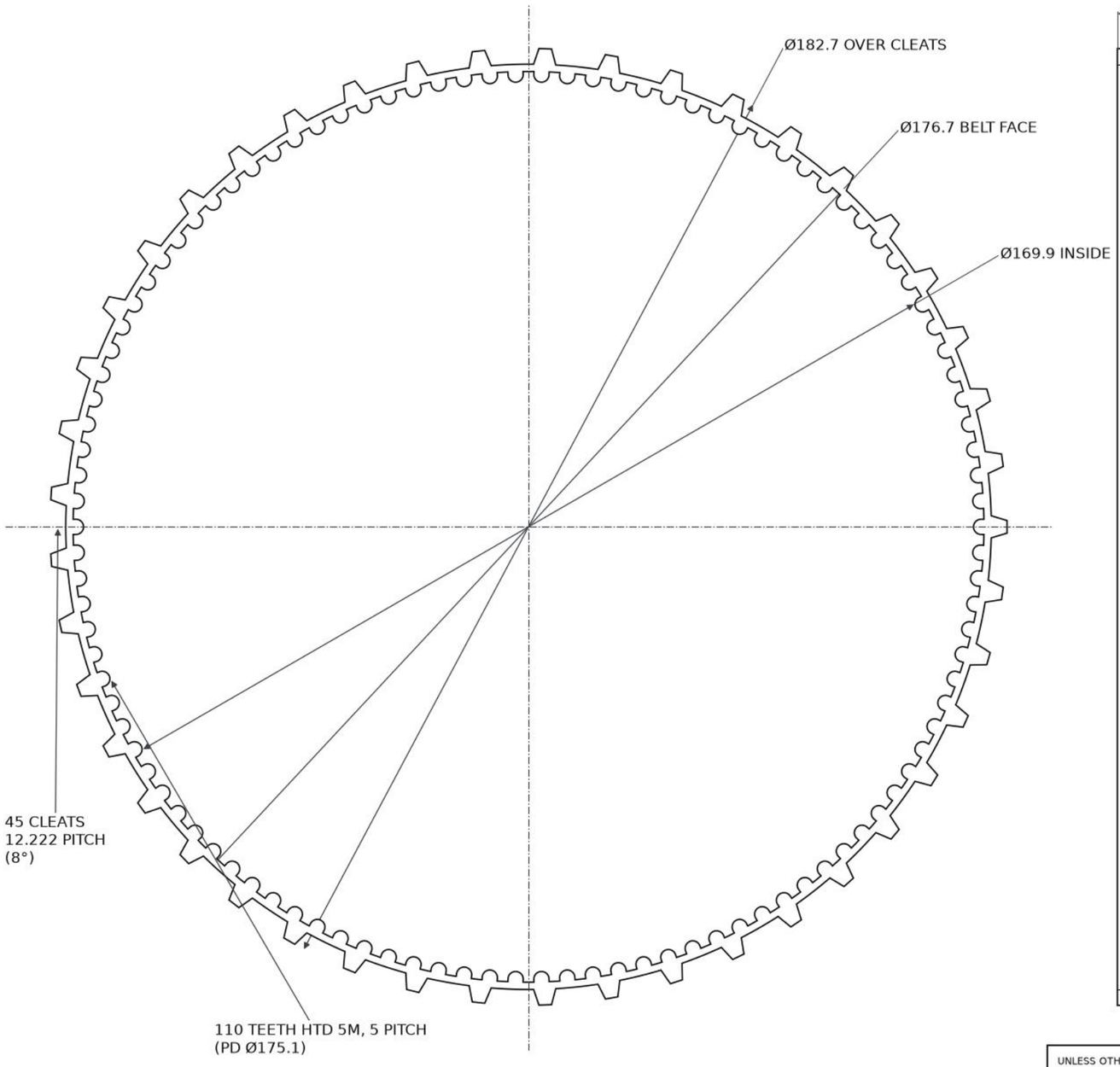
## 11 End view

Looking from the front: box in the middle (dashed), then each side a plate, a toothed wheel with its track, and the hub. 194 over the tracks, 224 over the plates.

## 12 Parts list

Every balloon number on the drawing matches a row here: part number, description and how many for this segment.

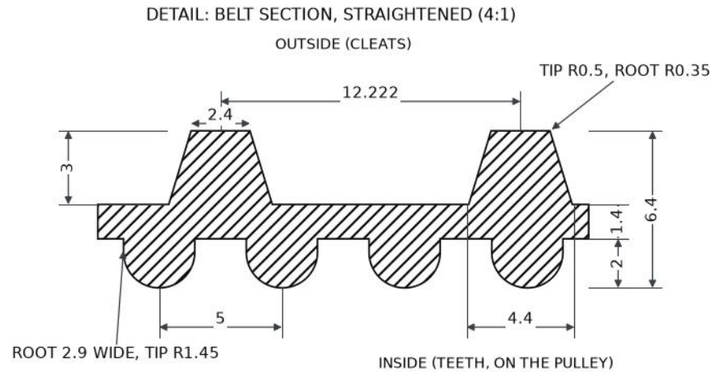
FRONT VIEW




EDGE VIEW




DETAIL: BELT SECTION, STRAIGHTENED (4:1)




NOTES:
1. FILE stl/track_TPU_x4.stl. PRINT STANDING AS A RING, NO SUPPORTS, 0.2 LAYERS, 100% INFILL, 20-30 MM/S, DIRECT-DRIVE EXTRUDER.
2. LOOP 550 ON THE PITCH LINE = 110 TEETH x 5 PITCH; RUNS ON TWO HTD 5M 20T PULLEYS (TS-06) AT 225 CENTRES. ABOUT 32 G.
3. 45 CLEATS AT 12.222 PITCH: 5 CLEATS = 61.11, ONE ES1 STEP ALONG THE SLOPE (35 x 50 AT 1:5), SO EVERY STEP EDGE SITS BETWEEN TWO CLEATS.
4. BED SMALLER THAN 190 MM: ASK FOR THE TWO-PIECE TRACK. PRINT THE 5 MM FIT-TEST SLICE FIRST (stl/fit_test_track_5mm_TPU.stl).

THIRD ANGLE



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**TRACK, 110T HTD 5M, 45 CLEATS**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: TPU 95A, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-01

QTY.: 4

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

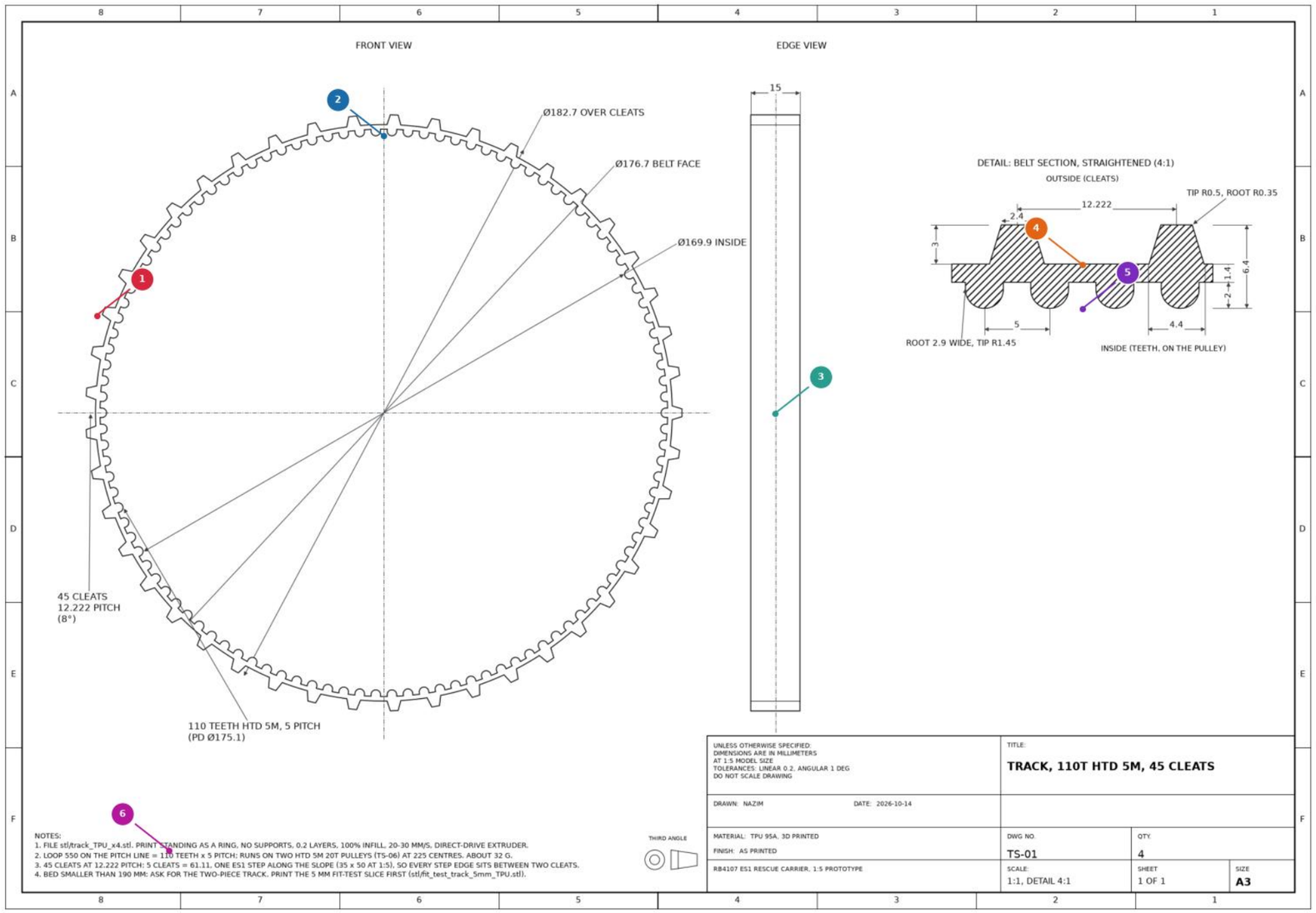
SCALE: 1:1, DETAIL 4:1

SHEET: 1 OF 1

SIZE: **A3**

# How to read TS-01: the track

Markers below sit on a reduced copy of sheet TS-01 (previous page, full size). Numbers match the key on the right.




**1 Cleats, outside**

45 bumps 12.22 apart (8 degrees). Five of them span one ES1 step, so every step edge lands between two cleats and the track holds like a gear on a rack.

**2 Teeth, inside**

110 HTD 5M teeth, 5 apart. These mesh with the 20-tooth wheels; this is what stops the track slipping.

**3 Edge view**

The ring seen from the side: 15 wide, 182.7 tall over the cleats.

**4 Cleat profile (4:1)**

A cleat is 3 tall, 4.4 wide at its base and 2.4 at the tip, with rounded corners so it releases from the step.

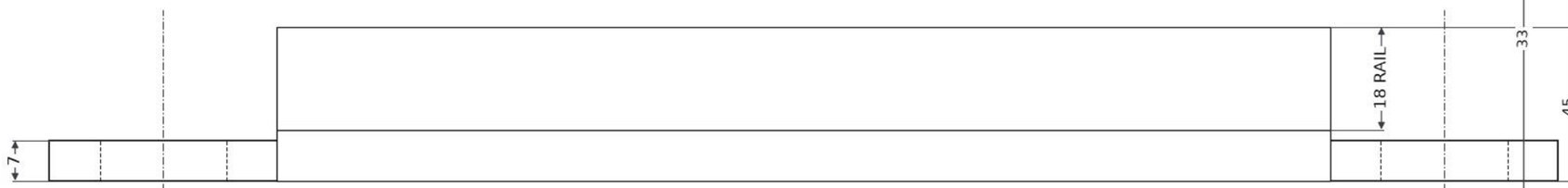
**5 Tooth profile (4:1)**

A tooth is 2 deep with a 1.45 radius tip and a 2.9 root; the 1.4 backing between teeth and cleats is the belt itself. Whole section 6.4 thick.

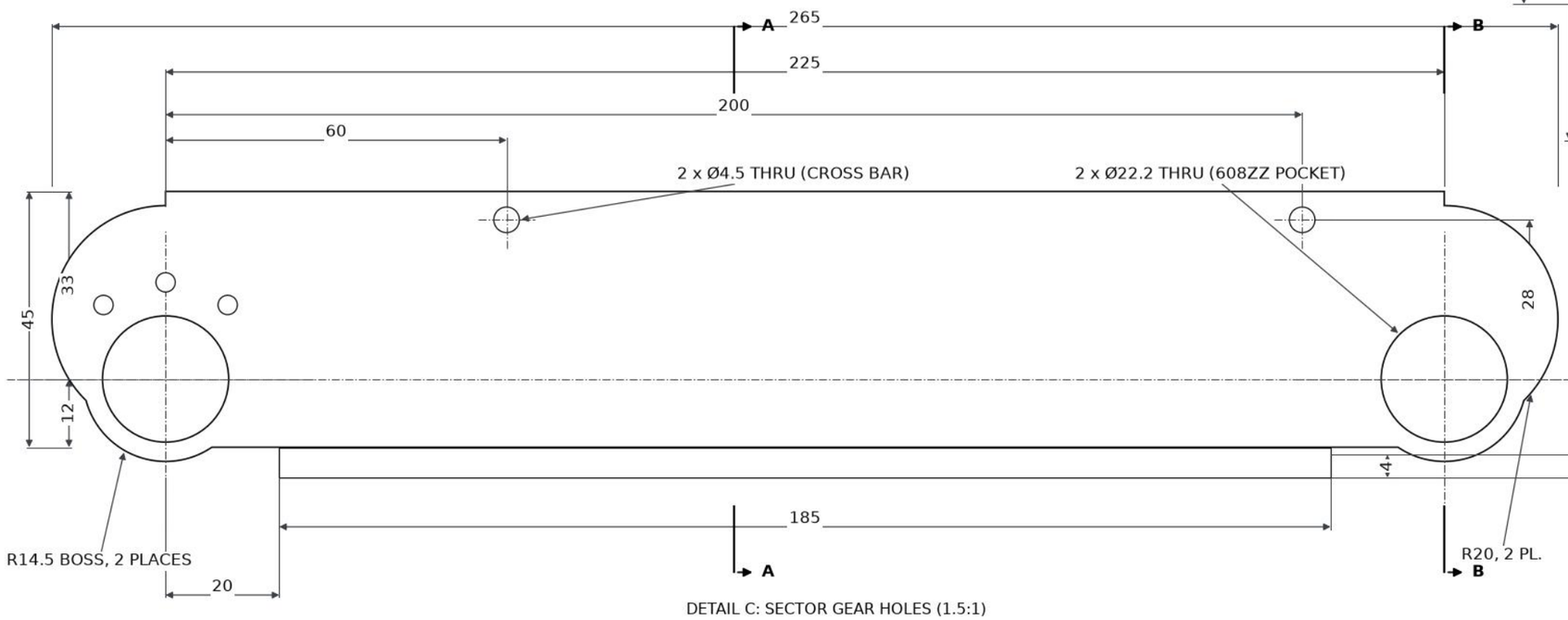
**6 Notes**

Print standing up as a ring, no supports, 100% infill, TPU 95A, on the Fab Lab UP300. Four tracks, about 32 g each.

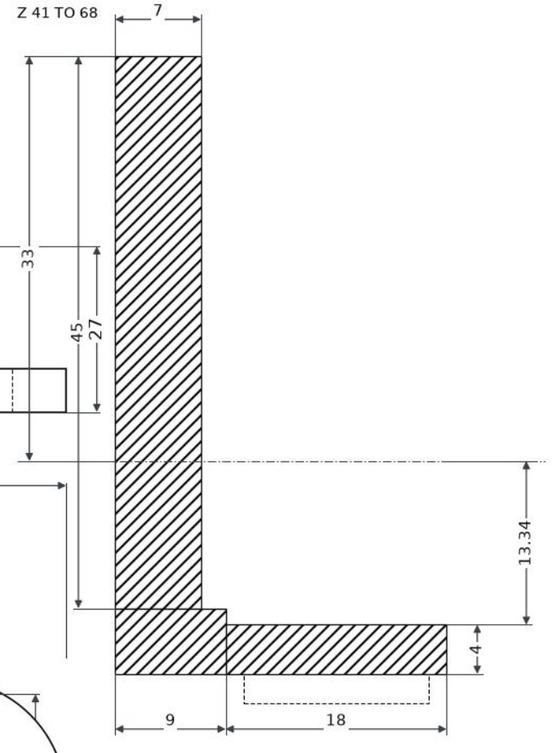
TOP VIEW




FRONT VIEW (FROM OUTBOARD)

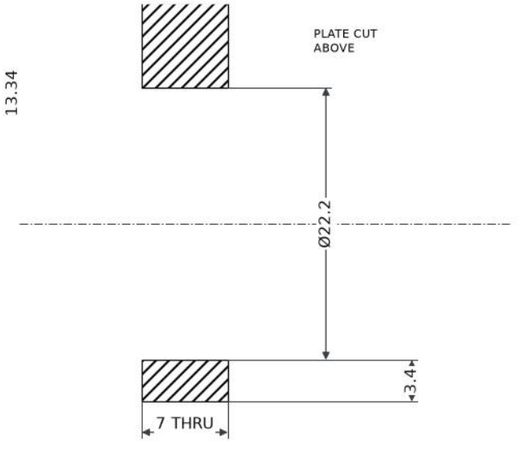



SECTION A-A (2:1)

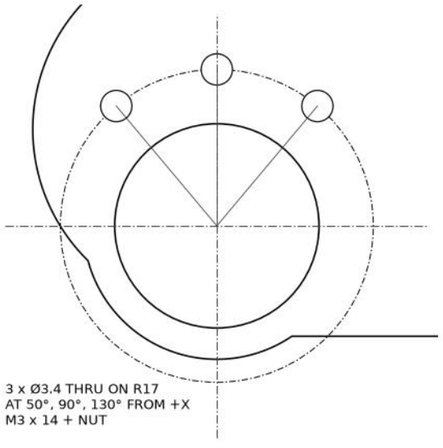



TRACK LOWER RUN (REF)

SECTION B-B (2:1)




DETAIL C: SECTOR GEAR HOLES (1.5:1)



3 x Ø3.4 THRU ON R17
AT 50°, 90°, 130° FROM +X
M3 x 14 + NUT

THIRD ANGLE

NOTES:
1. LEFT PLATE SHOWN; THE RIGHT ONE IS ITS MIRROR. FILES stl/side_plate_front_L.stl, _R.stl.
2. PRINT LYING ON THE PLATE FACE, 0.2 LAYERS, 4 WALLS, 30% INFILL, NO SUPPORTS.
3. BED 235 MM (OR 225 DIAGONAL); ON A SMALLER BED ASK FOR THE TWO-PIECE PLATE.
4. Z FROM THE VEHICLE CENTRE LINE: PLATE 41-48, WEB 41-50, RAIL 50-68
5. (RAIL CENTRED UNDER THE TRACK LANE 49-69). R20 ENDS ARE CENTRED 10.5 ABOVE THE AXLE.
6. POCKETS Ø22.2 THRU TAKE A 608ZZ (TS-07), LIGHT PRESS: MEASURE THE FIRST PRINT,
7. REAM IF TIGHT, GLUE IF LOOSE. THE RAIL TOP CARRIES THE LOWER RUN'S TOOTH TIPS.
8. X = 0 IS THE MIDDLE AXLE (HINGE); 3 x M3 SECTOR GEAR HOLES ON R17 ROUND IT, INBOARD FACE.

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:

**SIDE PLATE, FRONT SEGMENT**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: PETG, 3D PRINTED
FINISH: AS PRINTED

DWG NO. TS-02

QTY. 2 (1 L + 1 R)

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

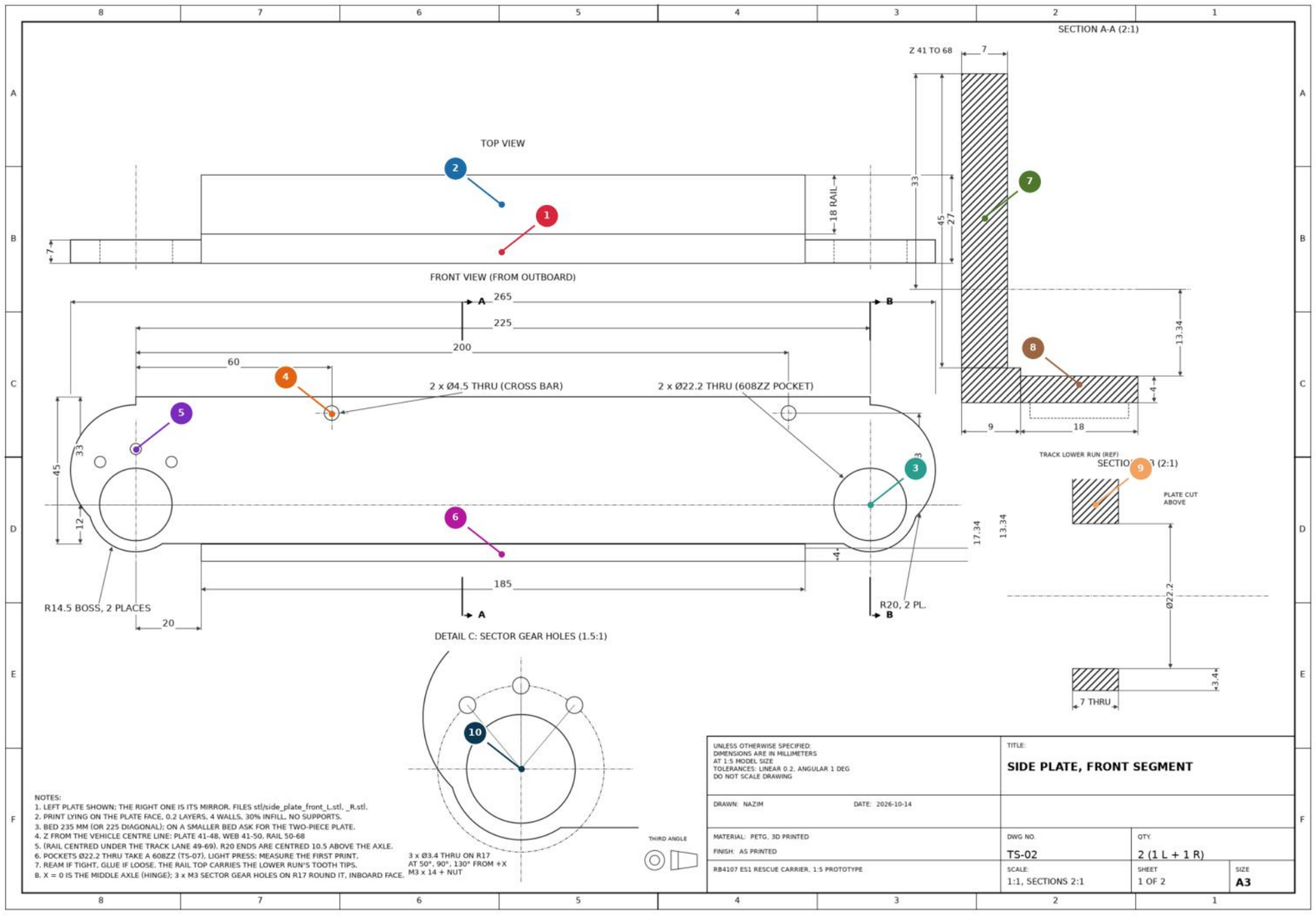
SCALE: 1:1, SECTIONS 2:1

SHEET 1 OF 2

SIZE **A3**

# How to read TS-02: the front segment's side plate

Markers below sit on a reduced copy of sheet TS-02 (previous page, full size). Numbers match the key on the right.




**1 Plate, seen from above**

7 thick, 265 long, 45 tall. Printed flat on the bed in PETG. Left plate drawn; the right one is its mirror.

**2 Slide rail, seen from above**

An 18 wide, 4 thick strip 185 long, joined to the plate by a web. It sits under the track's lower run so the track cannot sag between the wheels.

**3 Bearing pocket**

Ø22.2 hole, one at each axle. A 608ZZ bearing is pressed in; the axle turns in it. The round boss round it keeps a 3.4 wall.

**4 Cross bar hole**

Ø4.5, two per plate at 60 and 200 from the axle, 28 above it. The M4 rods go through here.

**5 Sector gear holes (front plate only)**

Three Ø3.4 holes on R17 round the middle axle. The hinge sector gear bolts here with M3 x 14 and nuts, on the inboard face.

**6 Rail from the side**

The strip hanging below the plate between x 20 and 205. Its top is 13.34 below the axle centre: exactly where the track's tooth tips run.

**7 Section A-A: plate**

Cut through the rail at x 100 and looking along the plate: the 7 thick plate standing 45 tall.

**8 Section A-A: rail and web**

The rail (18 x 4) and the 9 web joining it to the plate, with the track's lower run dashed on top of it.

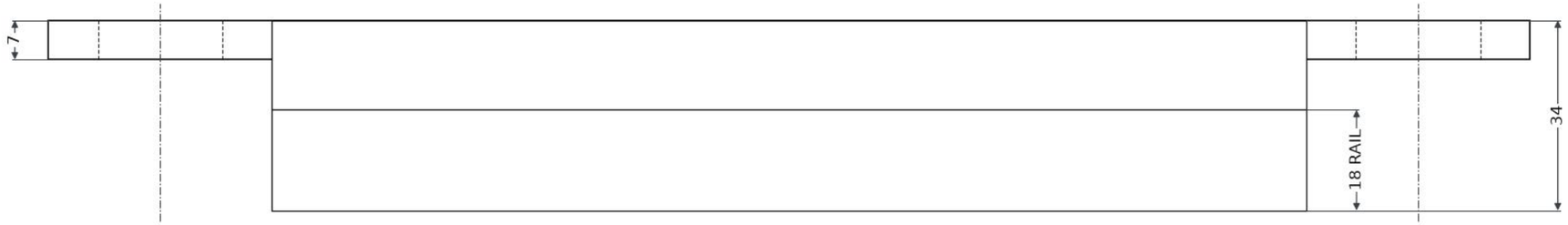
**9 Section B-B: pocket**

Cut through a bearing pocket: 7 thick, Ø22.2 straight through, 3.4 of wall round it.

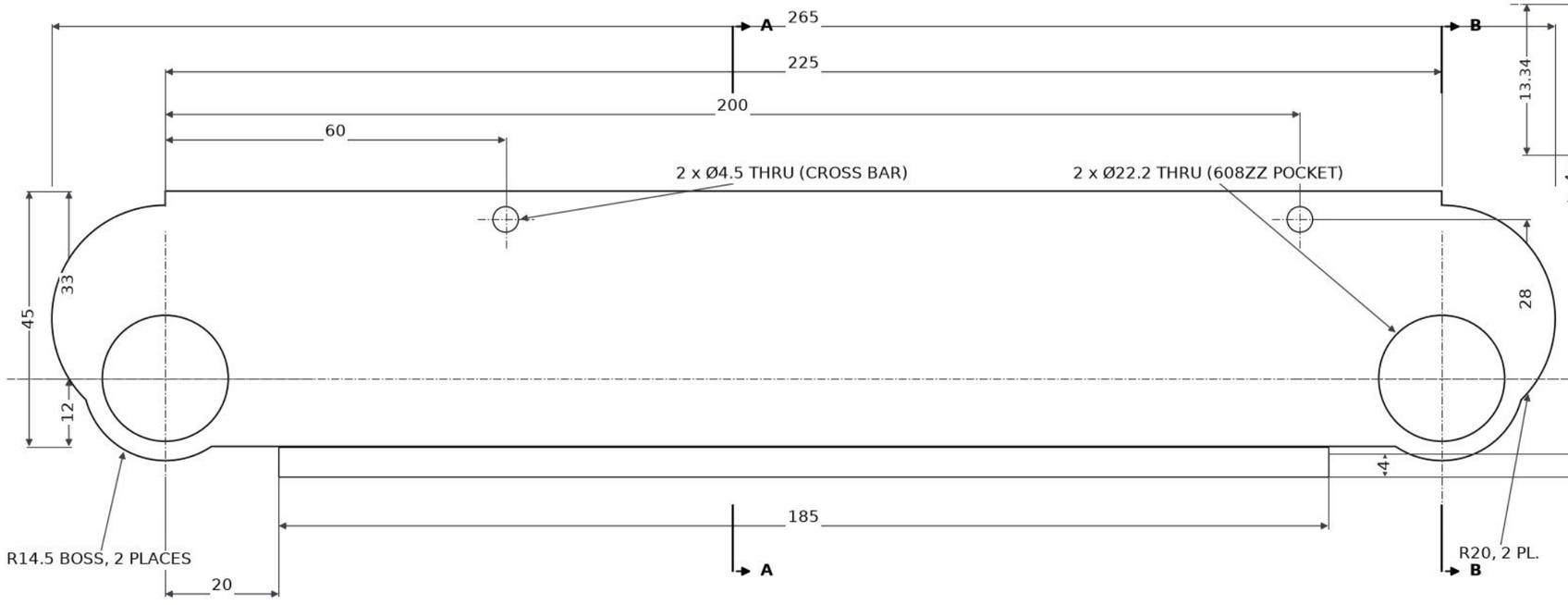
**10 Detail C**

The three sector gear holes at 50, 90 and 130 degrees from the forward direction, 1.5 times size.

TOP VIEW

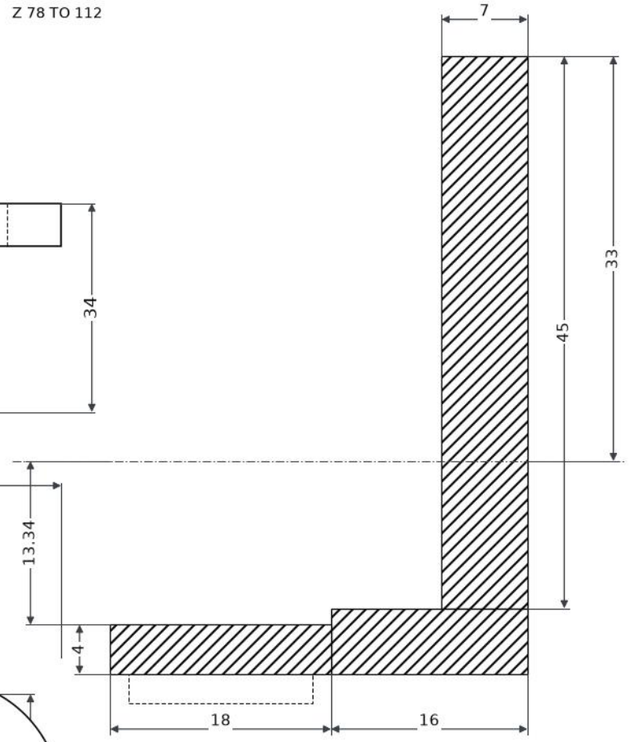



FRONT VIEW (FROM OUTBOARD)



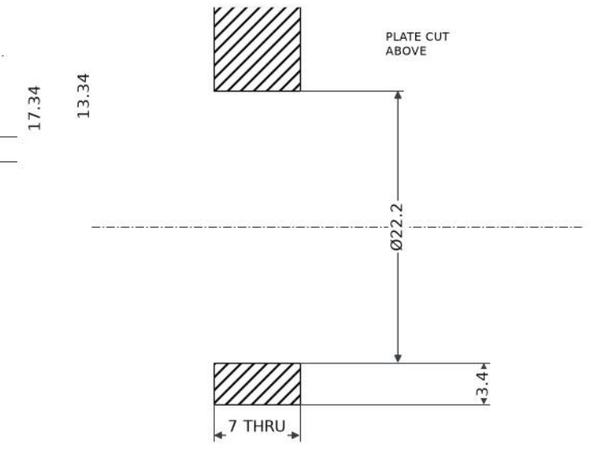

SECTION A-A (2:1)

Z 78 TO 112




TRACK LOWER RUN (REF)

SECTION B-B (2:1)




NOTES:
1. LEFT PLATE SHOWN; THE RIGHT ONE IS ITS MIRROR. FILES stl/side_plate_rear_L.stl, _R.stl.
2. PRINT LYING ON THE PLATE FACE, 0.2 LAYERS, 4 WALLS, 30% INFILL, NO SUPPORTS.
3. BED 235 MM (OR 225 DIAGONAL); ON A SMALLER BED ASK FOR THE TWO-PIECE PLATE.
4. Z FROM THE VEHICLE CENTRE LINE: PLATE 105-112, WEB 96-112, RAIL 78-96
5. (RAIL CENTRED UNDER THE TRACK LANE 77-97). R20 ENDS ARE CENTRED 10.5 ABOVE THE AXLE.
6. POCKETS Ø22.2 THRU TAKE A 608ZZ (TS-07), LIGHT PRESS: MEASURE THE FIRST PRINT,
7. REAM IF TIGHT, GLUE IF LOOSE. THE RAIL TOP CARRIES THE LOWER RUN'S TOOTH TIPS.
8. X = 0 IS THE REAR AXLE. THE BOX BOLTS TO THESE PLATES THROUGH THE Ø4.5 CROSS-BAR HOLES.

THIRD ANGLE



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**SIDE PLATE, REAR SEGMENT**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: PETG, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-03

QTY.: 2 (1 L + 1 R)

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

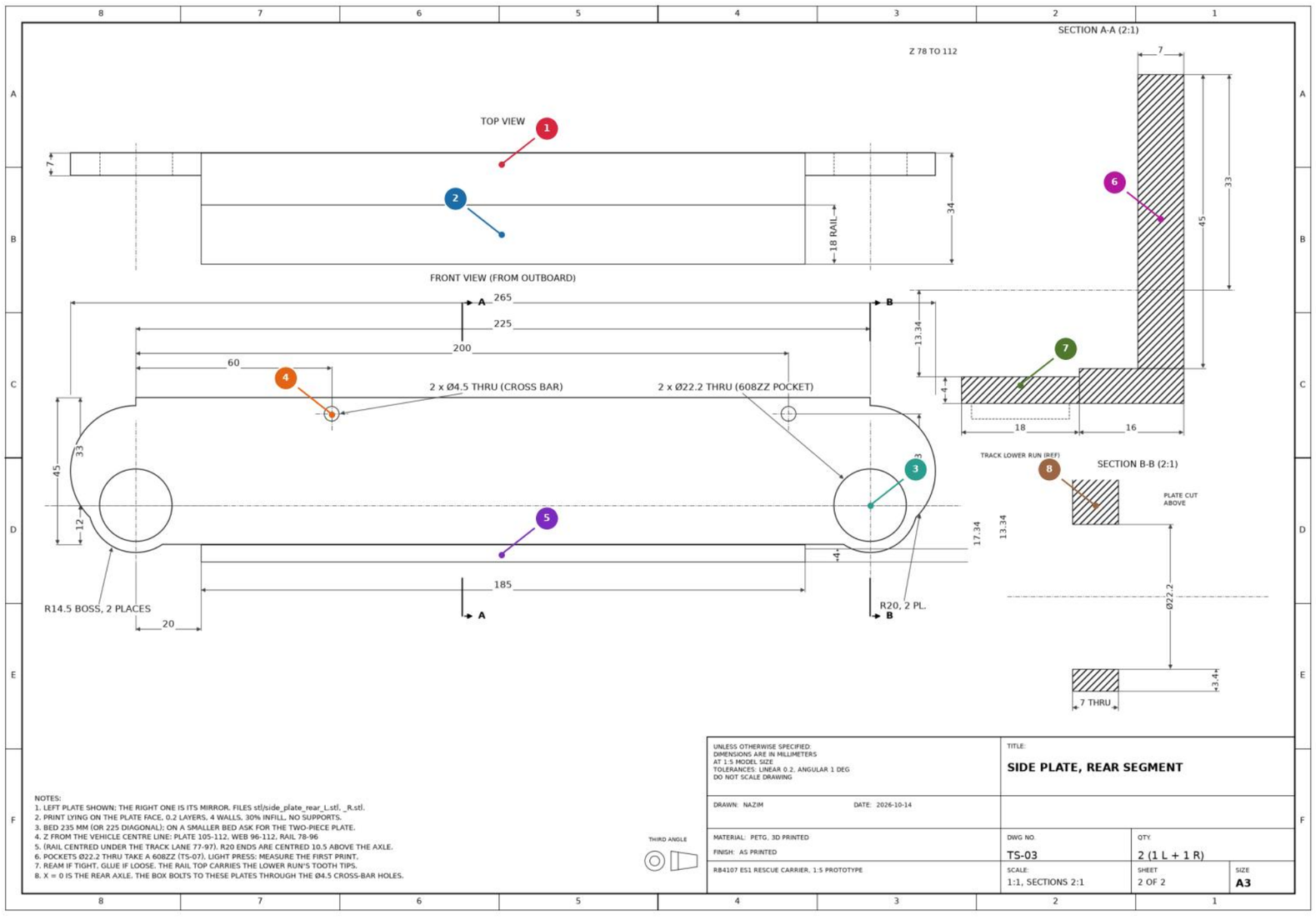
SCALE: 1:1, SECTIONS 2:1

SHEET: 2 OF 2

SIZE: **A3**

# How to read TS-03: the rear segment's side plate

Markers below sit on a reduced copy of sheet TS-03 (previous page, full size). Numbers match the key on the right.




**1 Plate, seen from above**

Same plate as TS-02 but on the outside of the vehicle (z 105 to 112), so the rail is on its inboard face and it has no sector gear holes.

**2 Slide rail, inboard**

The rail sits under the rear track lane (z 78 to 96), joined to the plate by a 16 wide web that passes under the pulley hub.

**3 Bearing pocket**

Ø22.2, a 608ZZ is pressed in at each axle (rear axle and middle axle).

**4 Cross bar hole**

Ø4.5 at 60 and 200 from the rear axle, 28 up. The M4 rods through these also clamp the box: plate, spacer, box wall, box wall, spacer, plate (sheet TS-08).

**5 Rail from the side**

Hangs below the plate between x 20 and 205; its top carries the lower run's tooth tips.

**6 Section A-A: plate**

The 7 thick plate, 45 tall, now on the right of the section because it is outboard.

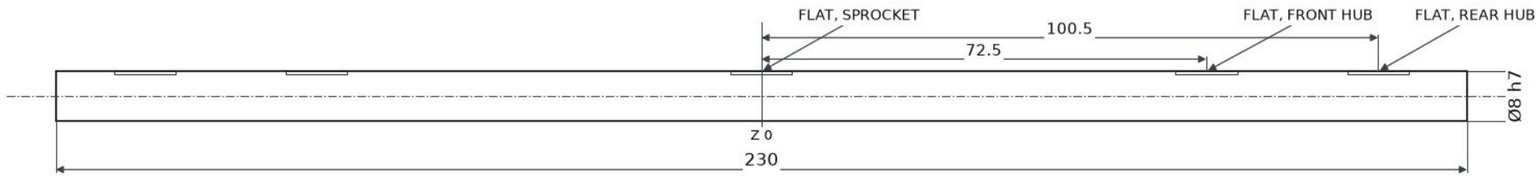
**7 Section A-A: rail**

18 x 4 rail with the track's lower run dashed on it, and the 16 web to the plate.

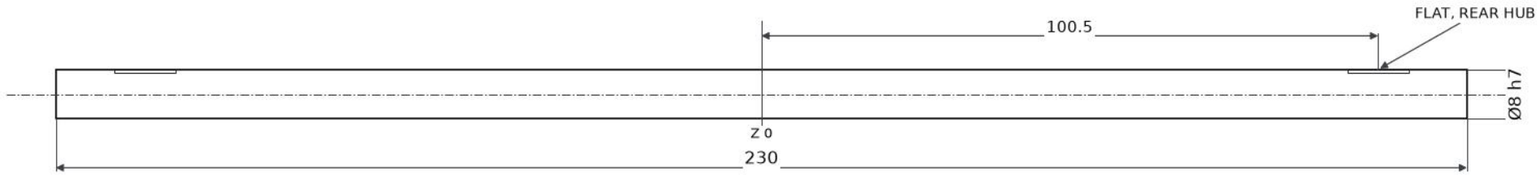
**8 Section B-B: pocket**

Cut through a pocket: Ø22.2 through the 7 plate, 3.4 wall.

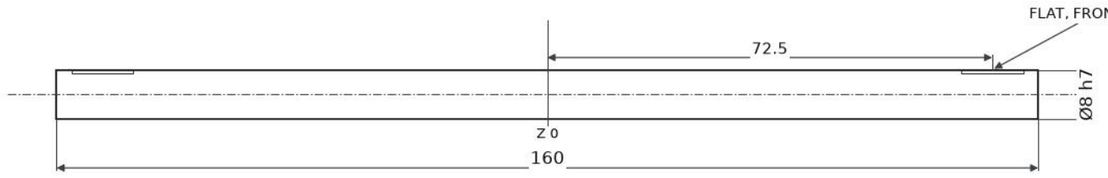
MIDDLE AXLE (HINGE, DRIVEN), QTY 1



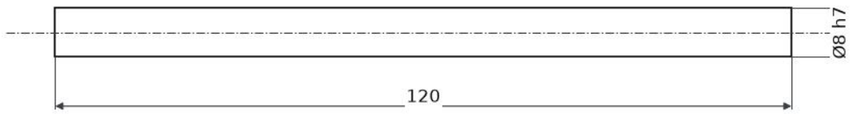

REAR AXLE, QTY 1



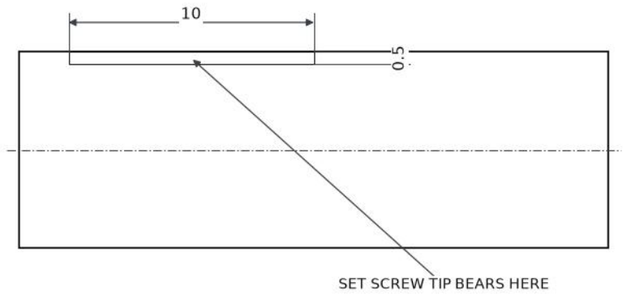

FRONT AXLE, QTY 1




HINGE CROSS SHAFT, QTY 1




DETAIL: FLAT UNDER A SET SCREW (4:1)




NOTES:
1. BUY: MISUMI 8 MM LINEAR SHAFT (UNHARDENED, SO THE FLATS CAN BE FILED) OR DESIGN LAB SPARE ROD. CUT TO LENGTH, ENDS SQUARE, DEBURRED: THE 608ZZ BORES ARE A CLOSE FIT.
2. FLATS 0.5 DEEP x 10 LONG UNDER EVERY PULLEY SET SCREW (AT THE HUB CENTRES, BOTH SIDES) AND UNDER THE DRIVE SPROCKET ON THE MIDDLE AXLE (Z 0).
3. THE MIDDLE AXLE IS THE HINGE: BOTH SEGMENTS' PLATES TURN ON IT IN THEIR 608ZZ. Z = 0 IS THE VEHICLE CENTRE LINE; THE AXLES ARE SYMMETRICAL ABOUT IT.
4. HINGE CROSS SHAFT: SAME ROD, 120 LONG, CARRIES THE TWO HINGE PINIONS ACROSS THE BOX IN 2 x 608ZZ; LENGTH TO BE CONFIRMED WITH THE HINGE GEAR DRAWING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**AXLES AND HINGE CROSS SHAFT**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: STEEL ROD Ø8 h7, UNHARDENED
FINISH: CUT SQUARE, DEBURR

DWG NO.: TS-04

QTY.: 3 + 1

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

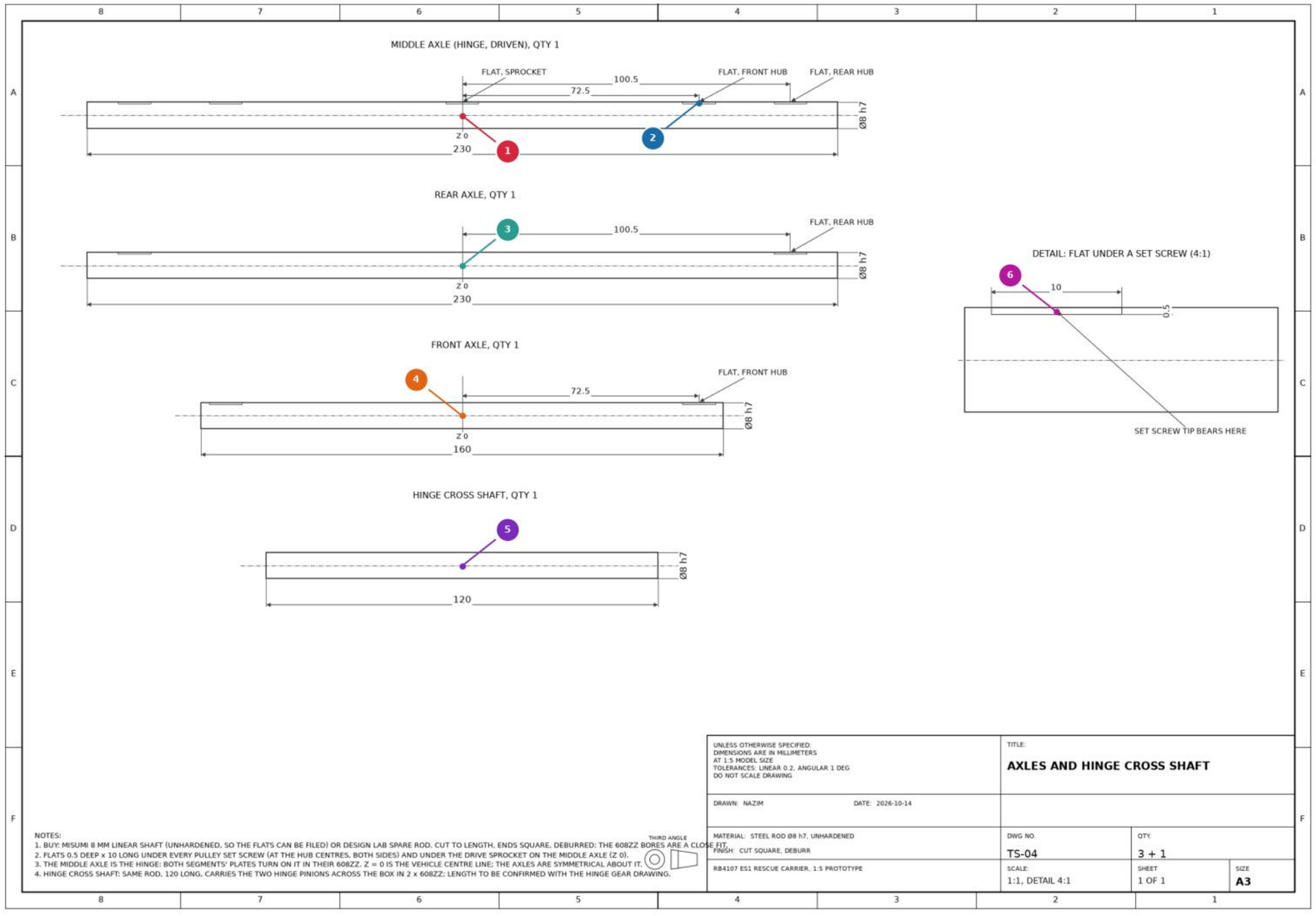
SCALE: 1:1, DETAIL 4:1

SHEET: 1 OF 1

SIZE: **A3**

# How to read TS-04: the axles

Markers below sit on a reduced copy of sheet TS-04 (previous page, full size). Numbers match the key on the right.




**1 Middle axle, 230**

The hinge shaft. Each side, from the centre out: front plate bearing, front segment wheel, rear segment wheel, rear plate bearing. Flats at z 0 (drive sprocket) and at both hubs each side.

**2 Flat**

A 0.5 deep, 10 long flat filed where each set screw lands, so the screw cannot slip round the shaft.

**3 Rear axle, 230**

Only the rear segment's two wheels and two plates. Flats at the rear hubs.

**4 Front axle, 160**

Only the front segment's two wheels and two plates. Shorter because the front plates are inboard.

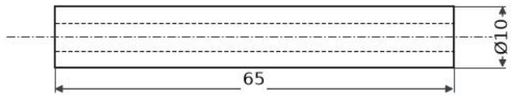
**5 Hinge cross shaft, 120**

Same rod. Runs across the box in two spare 608ZZ and carries the hinge pinions (team's part).

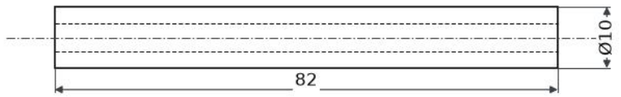
**6 Flat detail (4:1)**

What the flat looks like: 0.5 deep, 10 long. The set screw tip bears on the flat face.

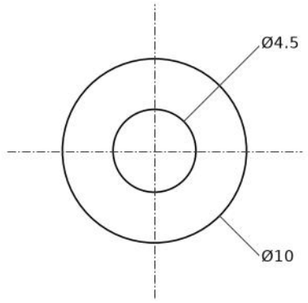
REAR SEGMENT SPACER, QTY 4 (2 PER ROD)



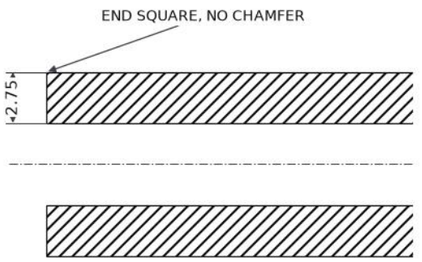

FRONT SEGMENT SPACER, QTY 2




END VIEW (3:1)




SECTION, ONE END (3:1)




NOTES:
1. FILES stl/spacer_front_x2.stl (82) AND stl/spacer_rear_x4.stl (65). PRINT STANDING ON END, 0.2 LAYERS, 3 WALLS, 20% INFILL.
2. FRONT: ONE 82 SPACER PER ROD BETWEEN THE FRONT PLATES (2 x 41). REAR: TWO 65 SPACERS PER ROD, ONE EACH SIDE BETWEEN THE BOX WALL (Z 40) AND THE REAR PLATE (Z 105).
3. THE NUTS OUTSIDE THE PLATES PULL PLATE - SPACER - BOX - SPACER - PLATE TOGETHER (SEE TS-08). AT 28 ABOVE THE AXLE THE SPACER CLEARS THE UPPER RUN'S CLEAT TIPS BY 3.3.
4. BORE 4.5 IS A CLEARANCE FOR THE M4 ROD. ENDS SQUARE: THEY SET THE PLATE SPACING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**CROSS BAR SPACER**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: PETG, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-05

QTY.: 2 x 82 + 4 x 65

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 1:1, DETAIL 3:1

SHEET: 1 OF 1

SIZE: **A3**

# How to read TS-05: the cross bar spacers

Markers below sit on a reduced copy of sheet TS-05 (previous page, full size). Numbers match the key on the right.

**1 Rear spacer, 65 long, 4 off**

Two per rear rod, one each side between the box wall (z 40) and the rear plate (z 105). The box sits between them.

**2 Front spacer, 82 long, 2 off**

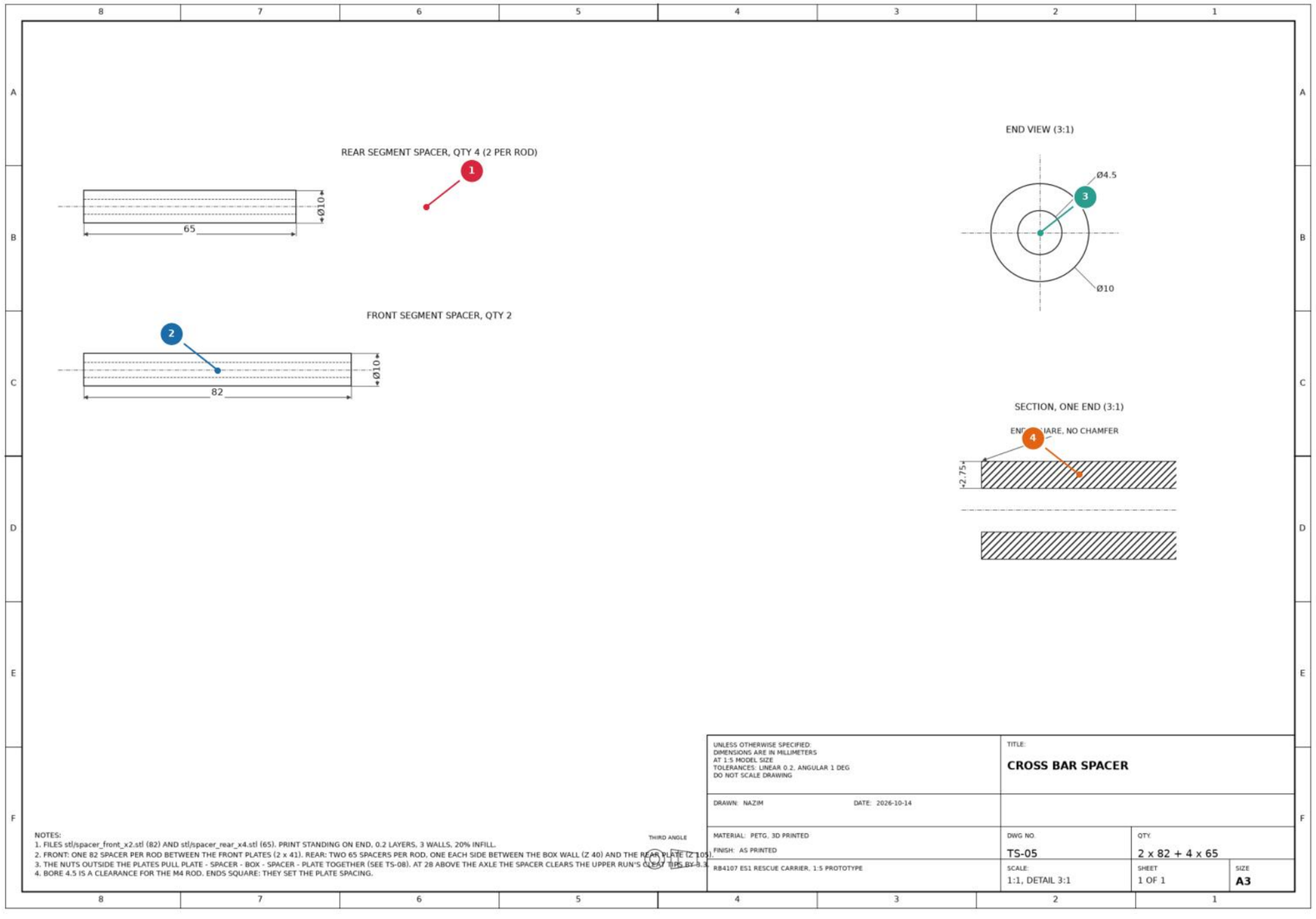
One per front rod, filling the whole gap between the two front plates (z -41 to 41).

**3 End view**

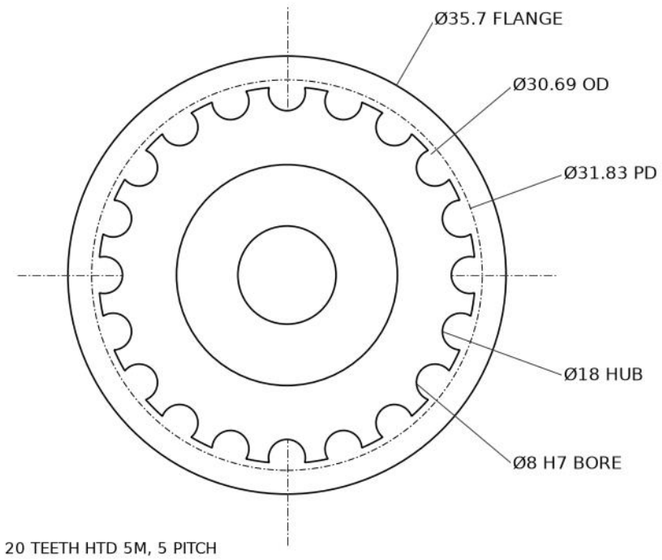
Ø10 outside, Ø4.5 bore: the M4 rod passes through with clearance.

**4 Section**

Just a tube, 2.75 wall, ends cut square because they set the plate spacing.

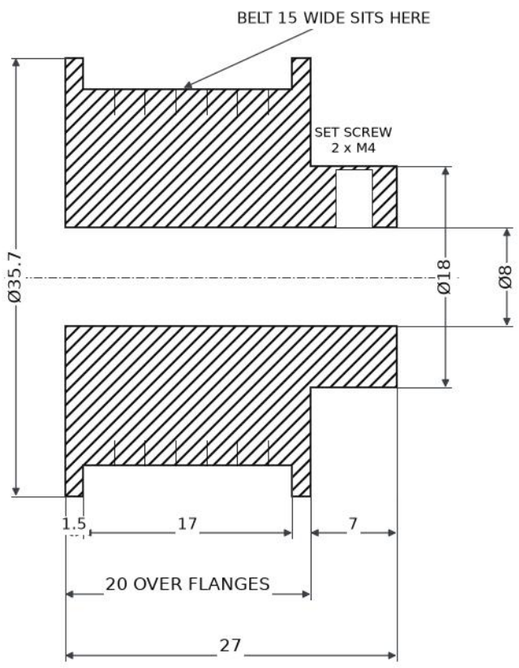


FRONT VIEW




20 TEETH HTD 5M, 5 PITCH

SECTION (ALONG THE AXLE)




NOTES:
1. BOUGHT PART. ORDER: HTD 5M, 20 TEETH, FOR 15 MM BELT, 8 MM BORE, FLANGES BOTH SIDES, SET-SCREW HUB. MISUMI SINGAPORE (APPROVED VENDOR). 4 PER SEGMENT.
2. WHAT MATTERS: 20T HTD 5M (PD 31.83), 15 MM BELT, 8 BORE. FLANGE, HUB AND LENGTH SHOWN ARE THE STAND-IN'S: CHECK THE VENDOR DRAWING.
3. LAYOUT ALLOWS 20 OVER THE FLANGES AND 7 FOR THE HUB (27 OVERALL) PER PULLEY. A LONGER PULLEY MOVES THE LANES: TELL THE TEAM BEFORE ORDERING.
4. PLA STAND-IN FOR FIT TESTS ONLY: stl/stand_in_pulley_20T_PLA_test_only.stl (M3 SET SCREW). NOT FOR DRIVING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**TIMING PULLEY HTD 5M 20T (BOUGHT)**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: ALUMINIUM (VENDOR'S)
FINISH: VENDOR'S

DWG NO.: TS-06

QTY.: 8

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

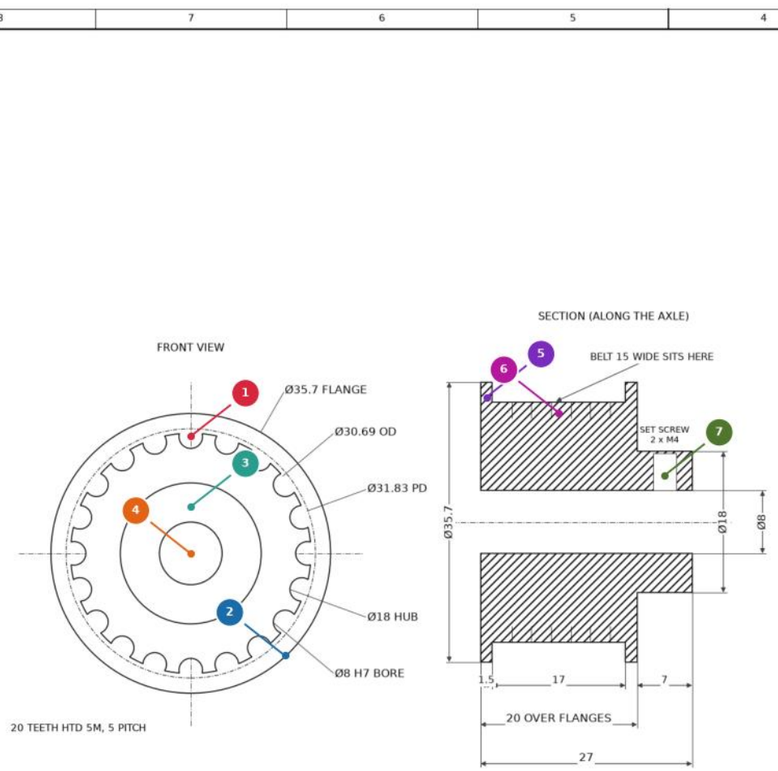
SCALE: 2:1

SHEET: 1 OF 1

SIZE: **A3**

# How to read TS-06: the toothed wheel (bought)

Markers below sit on a reduced copy of sheet TS-06 (previous page, full size). Numbers match the key on the right.




**1 — 20 grooves**

The track's teeth drop into these. HTD 5M profile, 5 pitch, pitch diameter 31.83.

**2 — Flange**

A thin Ø35.7 disc on each side of the teeth. Keeps the 15 wide track from walking off sideways.

**3 — Hub**

Ø18 boss, 7 long, on one side. The 2 M4 set screws sit in it and clamp the wheel to the axle.

**4 — Bore**

Ø8 H7: a close sliding fit on the Ø8 axle.

**5 — Flange, in section**

1.5 thick, both sides. Total 20 over the flanges, which is the lane width in the layout.

**6 — Belt seat**

17 between the flanges; the 15 wide track sits here with 1 each side.

**7 — Hub and set screws**

The 7 long hub with 2 x M4 set screws onto the axle flat. 27 overall length per wheel.

NOTES:
1. BOUGHT PART. ORDER: HTD 5M, 20 TEETH, FOR 15 MM BELT, 8 MM BORE, FLANGES BOTH SIDES, SET-SCREW HUB. MISUMI SINGAPORE (APPROVED VENDOR). 4 PER SEGMENT.
2. WHAT MATTERS: 20T HTD 5M (PD 31.83), 15 MM BELT, 8 BORE, FLANGE, HUB AND LENGTH SHOWN ARE THE STAND-IN'S: CHECK THE VENDOR DRAWING.
3. LAYOUT ALLOWS 20 OVER THE FLANGES AND 7 FOR THE HUB (27 OVERALL) PER PULLEY. A LONGER PULLEY MOVES THE LANES: TELL THE TEAM BEFORE ORDERING.
4. PLA STAND-IN FOR FIT TESTS ONLY: stl/stand_in_pulley_20T_PLA_test_only.stl (M3 SET SCREW). NOT FOR DRIVING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

DRAWN: NAZIM DATE: 2026-10-14

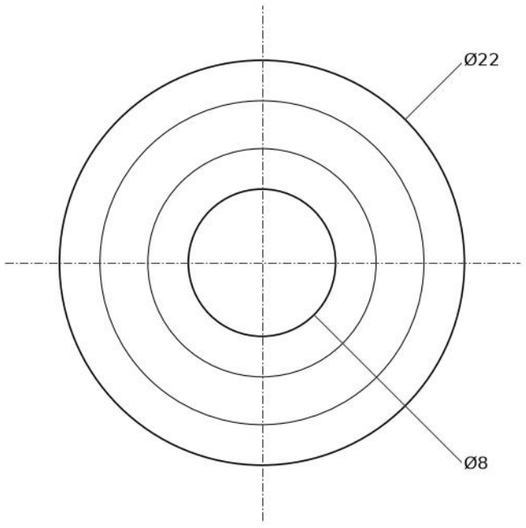
MATERIAL: ALUMINIUM (VENDOR'S)
FINISH: VENDOR'S

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

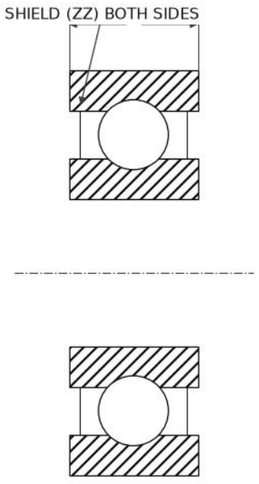
TITLE: **TIMING PULLEY HTD 5M 20T (BOUGHT)**

DWG NO. TS-06
QTY. 8
SCALE: 2:1
SHEET 1 OF 1
SIZE **A3**

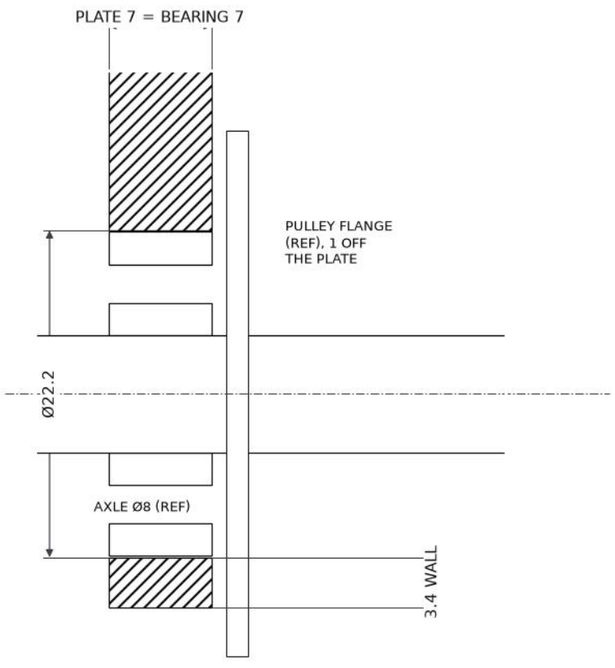
FRONT VIEW




SECTION




FITTED IN THE PLATE POCKET (2.4:1)




NOTES:
1. BOUGHT PART. 608ZZ, 8 x 22 x 7, SHIELDED BOTH SIDES. RS SINGAPORE (RS PRO 608-2Z) OR WHEELER MECTRADE / SKF (APPROVED VENDORS). ABOUT S$3 EACH.
2. QTY 10: 8 PRESSED INTO THE SIDE PLATE POCKETS (2 PER PLATE, TS-02 / TS-03), 2 IN THE BOX WALLS FOR THE HINGE CROSS SHAFT (TS-04).
3. POCKET Ø22.2 THRU IN THE PETG PLATE = OD + 0.2, A LIGHT PRESS. THE R14.5 BOSS GIVES A 3.4 WALL. MEASURE THE FIRST PRINT; REAM IF TIGHT, GLUE (CA) IF LOOSE.
4. REPLACES THE KFL08 FLANGE BLOCK ON THE QUOTATION SLIDE: 48 x 26 x 12 DOES NOT FIT IN THE 9 MM BETWEEN THE BOX (Z 40) AND THE FIRST TRACK (Z 49).

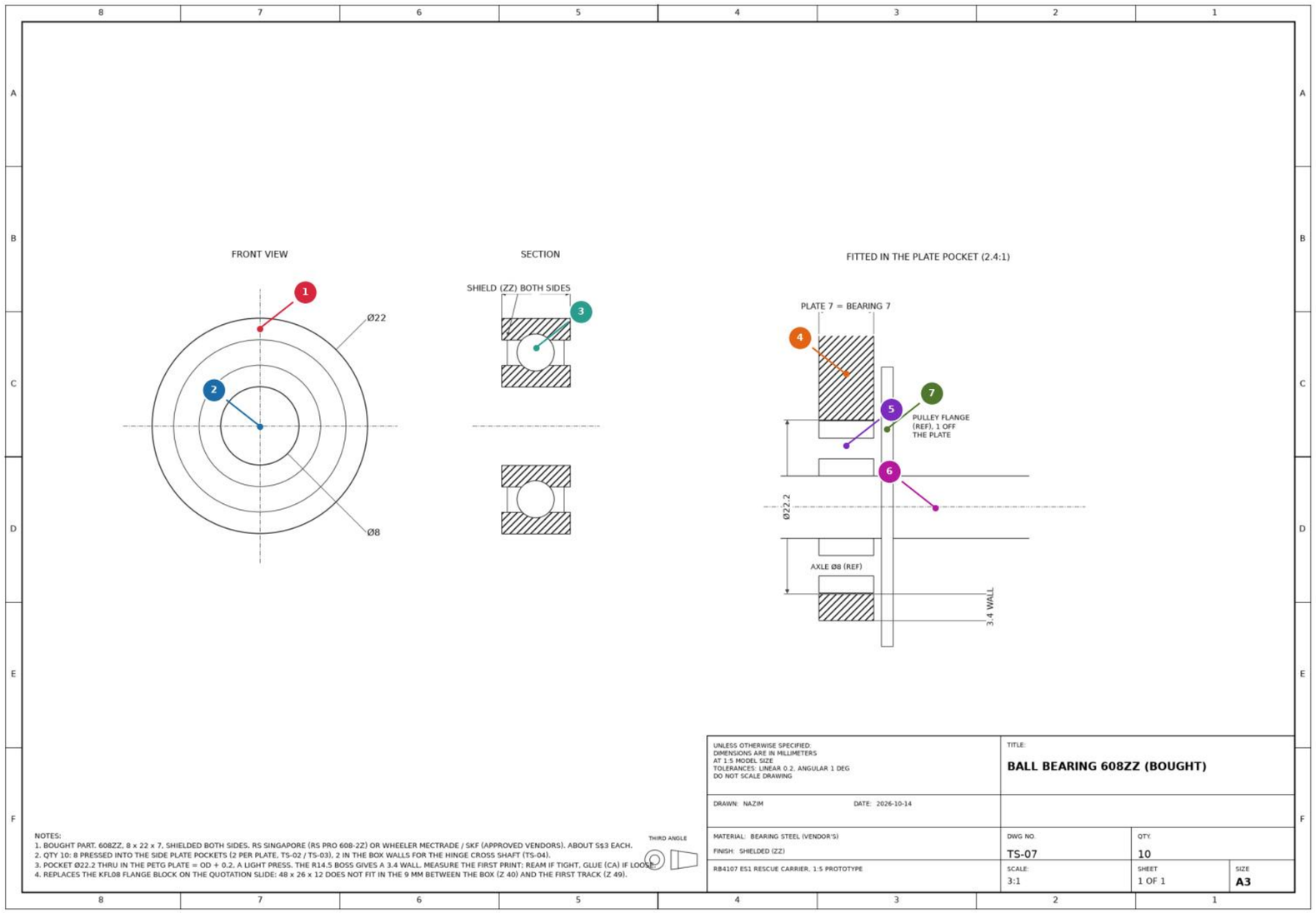
THIRD ANGLE



| | | | |
|---|---|---|---|
| UNLESS OTHERWISE SPECIFIED:<br>DIMENSIONS ARE IN MILLIMETERS<br>AT 1:5 MODEL SIZE<br>TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG<br>DO NOT SCALE DRAWING | TITLE:<br>**BALL BEARING 608ZZ (BOUGHT)** | | |
| DRAWN: NAZIM DATE: 2026-10-14 | | | |
| MATERIAL: BEARING STEEL (VENDOR'S)<br>FINISH: SHIELDED (ZZ) | DWG NO.<br>TS-07 | QTY.<br>10 | |
| RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE | SCALE:<br>3:1 | SHEET<br>1 OF 1 | SIZE<br>**A3** |

# How to read TS-07: the bearing (bought)

Markers below sit on a reduced copy of sheet TS-07 (previous page, full size). Numbers match the key on the right.




**1 Outer race, Ø22**

Pressed into the plate's Ø22.2 pocket. It does not turn.

**2 Bore, Ø8**

The axle slides through; the inner race turns with the axle.

**3 Balls**

Between the races, 7 wide in all. Shielded (ZZ) both sides so dust stays out.

**4 Plate**

The 7 PETG plate: plate thickness equals bearing width, so the bearing sits flush.

**5 Bearing in its pocket**

A light press fit (0.2 over). Measure the first print; ream if tight, glue if loose.

**6 Axle**

Ø8 rod through the bore, continuing to the wheel.

**7 Pulley flange next to it**

The wheel's flange starts 1 beyond the plate face, so the wheel never rubs the plate.

NOTES:
1. BOUGHT PART. 608ZZ, 8 x 22 x 7, SHIELDED BOTH SIDES. RS SINGAPORE (RS PRO 608-2Z) OR WHEELER MECTRADE / SKF (APPROVED VENDORS). ABOUT S$3 EACH.
2. QTY 10: 8 PRESSED INTO THE SIDE PLATE POCKETS (2 PER PLATE, TS-02 / TS-03), 2 IN THE BOX WALLS FOR THE HINGE CROSS SHAFT (TS-04).
3. POCKET Ø22.2 THRU IN THE PETG PLATE = OD + 0.2, A LIGHT PRESS. THE R14.5 BOSS GIVES A 3.4 WALL. MEASURE THE FIRST PRINT; REAM IF TIGHT, GLUE (CA) IF LOOSE.
4. REPLACES THE KFL08 FLANGE BLOCK ON THE QUOTATION SLIDE: 48 x 26 x 12 DOES NOT FIT IN THE 9 MM BETWEEN THE BOX (Z 40) AND THE FIRST TRACK (Z 49).

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

DRAWN: NAZIM DATE: 2026-10-14

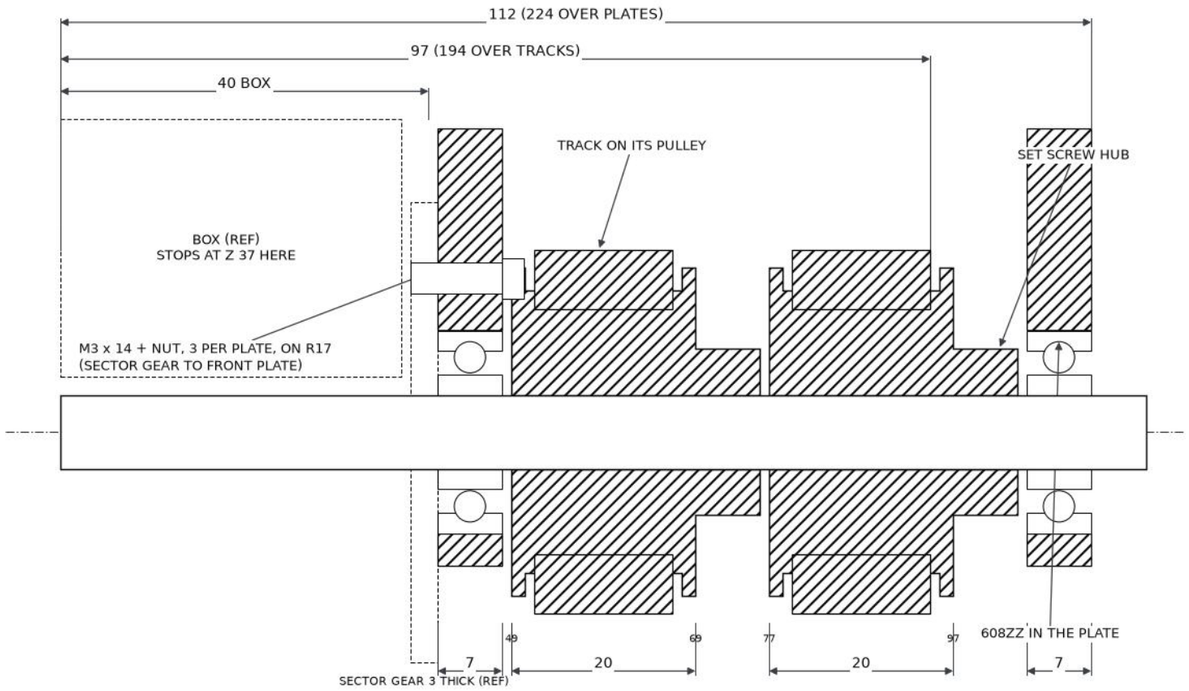
MATERIAL: BEARING STEEL (VENDOR'S)
FINISH: SHIELDED (ZZ)

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

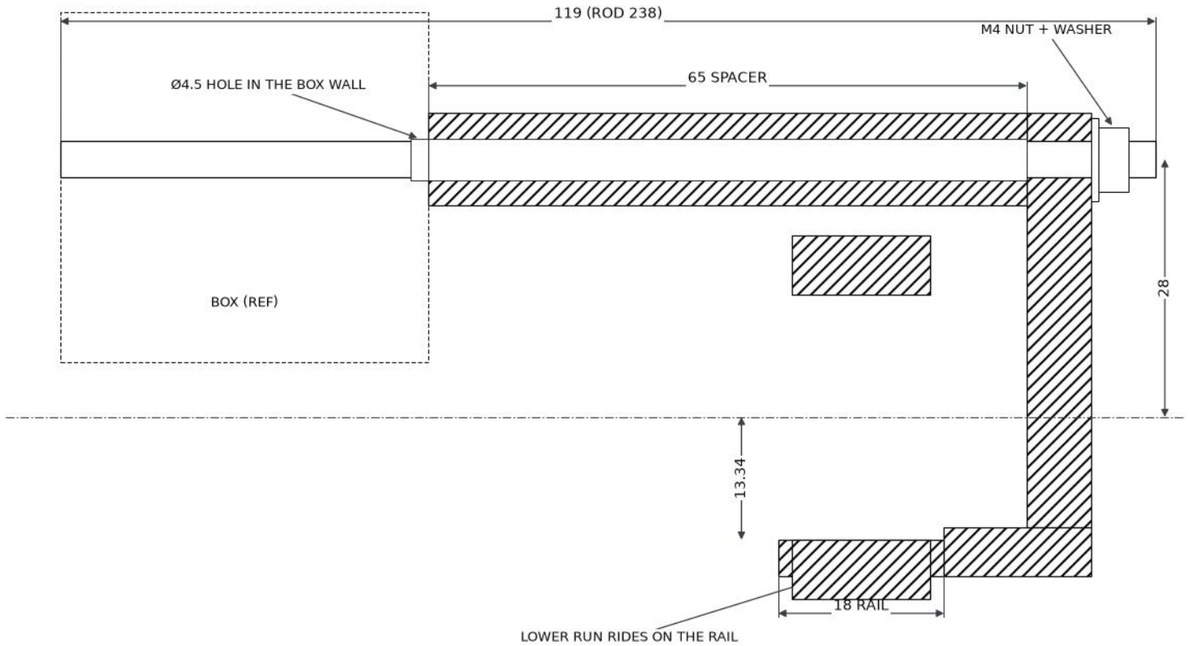
TITLE: **BALL BEARING 608ZZ (BOUGHT)**

| DWG NO. | QTY | |
|---|---|---|
| TS-07 | 10 | |
| SCALE: 3:1 | SHEET: 1 OF 1 | SIZE: **A3** |

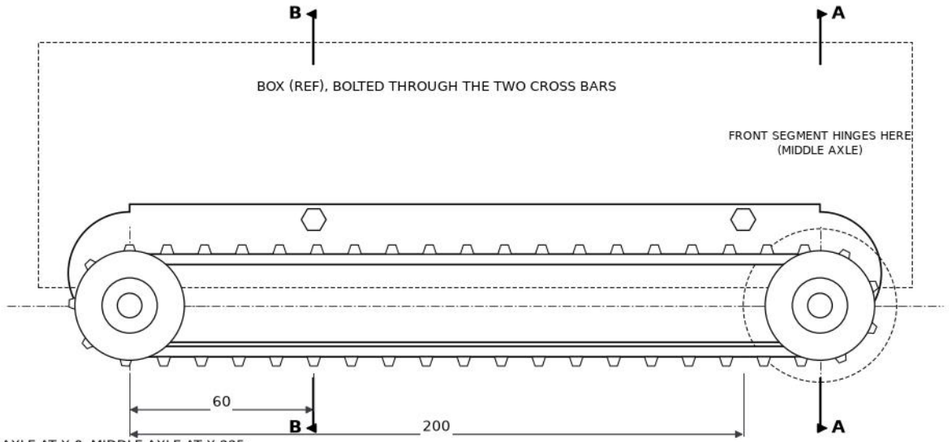
SECTION A-A THROUGH THE MIDDLE AXLE (HINGE), 1.5:1




SECTION B-B THROUGH A REAR CROSS BAR (X 60 FROM THE REAR AXLE), 1.5:1




SIDE VIEW, REAR SEGMENT ON THE BOX (FROM OUTBOARD), 1:2




NOTES:
1. ATTACHMENT: THE REAR SEGMENT BOLTS TO THE BOX BY ITS TWO CROSS-BAR RODS (M4 x 238), NUT + WASHER EACH END:
2. PLATE - 65 SPACER - BOX WALL - BOX WALL - 65 SPACER - PLATE. BOX NEEDS 2 x Ø4.5 HOLES PER WALL AT X 60 AND 200, 28 ABOVE THE AXLE.
3. THE FRONT SEGMENT IS HELD ONLY BY THE MIDDLE AXLE (HINGE) AND THE SECTOR GEARS, SO IT CAN SWING.
4. ONE SIDE SHOWN (Z FROM THE CENTRE LINE), THE OTHER MIRRORS. BOX, SECTOR GEAR AND PINION: TEAM'S PARTS, DASHED.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**REAR SEGMENT TO THE BOX**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: SEE TS-00
FINISH:

DWG NO. TS-08

QTY. 1

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

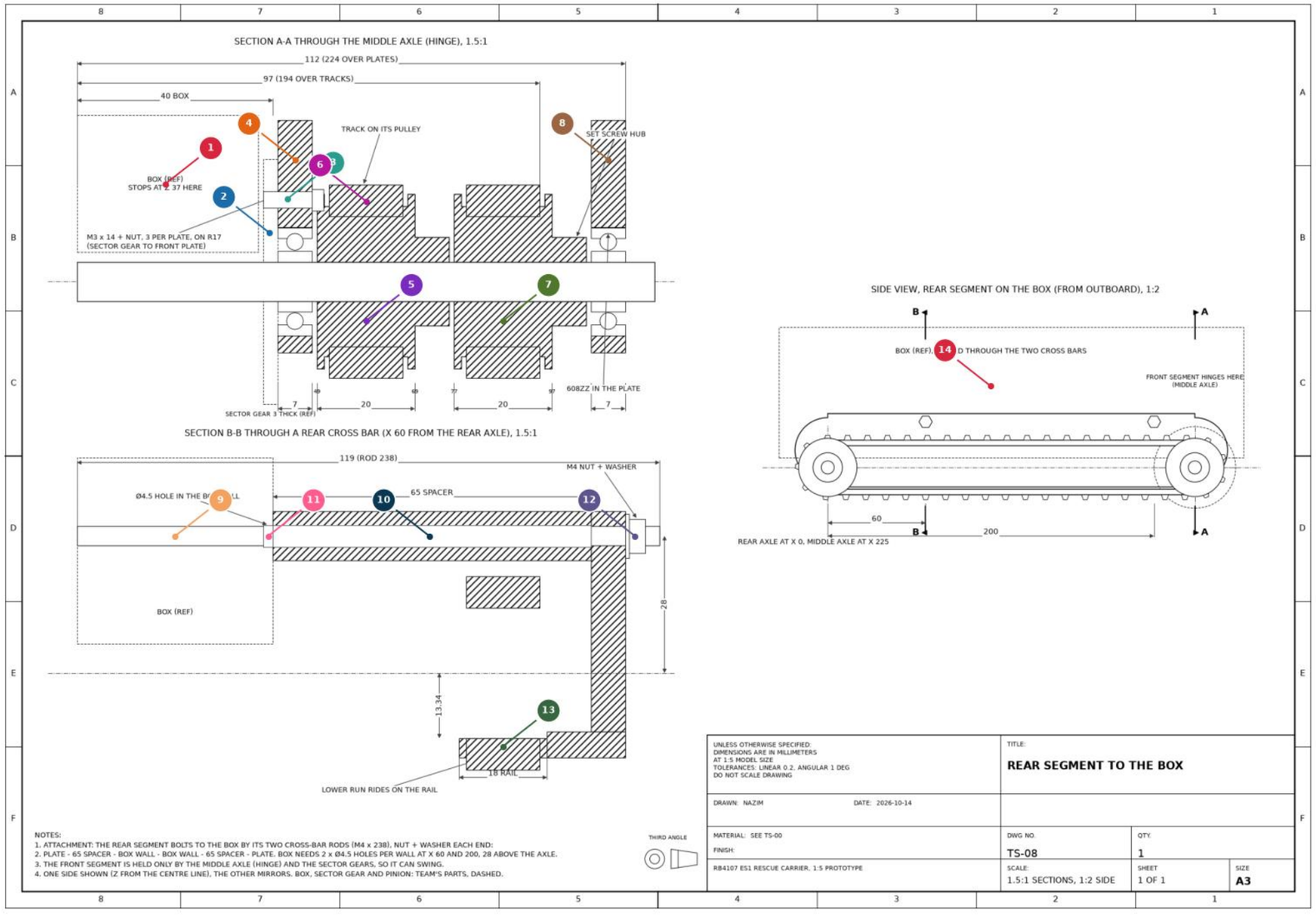
SCALE: 1.5:1 SECTIONS, 1:2 SIDE

SHEET 1 OF 1

SIZE **A3**

# How to read TS-08: how the rear segment hangs on the box

Markers below sit on a reduced copy of sheet TS-08 (previous page, full size). Numbers match the key on the right.




**1 — Box (dashed)**
The team's box, 80 wide. Near the middle axle it must stop at z 37 to leave 3 for the sector gear.

**2 — Hinge sector gear (dashed)**
3 thick, bolted to the front plate's inboard face. The pinion on the hinge cross shaft turns it to bend the front segment.

**3 — M3 x 14 bolt and nut**
One of three per plate, through the gear and the plate, nut on the outboard side.

**4 — Front plate**
7 thick with its 608ZZ; the middle axle turns in it.

**5 — Front segment wheel**
Set-screwed to the middle axle; its hub points outboard.

**6 — Front track**
Wrapped on that wheel, cut through.

**7 — Rear segment wheel and track**
Next outboard, same again. Both segments are driven by this one axle.

**8 — Rear plate**
Outermost, 105 to 112, with its 608ZZ. 224 over the two rear plates.

**9 — M4 rod, 238 long**
Runs plate to plate through the box. This is the only attachment of the rear segment to the box.

**10 — 65 spacer**
One each side between box wall and plate. The rod passes through it.

**11 — Ø4.5 hole in the box wall**
The box needs two of these per wall at x 60 and 200 from the rear axle, 28 above the axle line.

**12 — M4 nut and washer**
Tightened on each end: plate, spacer, box, spacer, plate are clamped as one.

**13 — Rail under the track**
The lower run of the rear track rests on the plate's rail; the upper run is free above.

**14 — Side view: box on the segment**
The box is bolted through the two cross bars (hex nuts visible). The front segment hinges on the middle axle at the right.

# Parts list and build guide: track segments of the 1:5 prototype

Numbers from `parts_spec.json` v2 and the TS-00 to TS-07 sheets. Prices S$ ex GST as seen in vendor listings on 14 Oct 2026 (vendor pages could not be opened from this machine: confirm at checkout).

## 1. Does the track fit the school printers? Yes, in one piece.

| Machine (NUS CDE) | Bed | Track ring Ø183 x 15, printed standing | Side plate 265 x 50 x 11, flat |
|---|---|---|---|
| Tiertime UP300, Fab Lab E4-02-06 (has a TPU extruder head) | 205 x 255 x 225 mm | **fits** (183 wide, 183 tall) | **fits on the diagonal** (327 mm available; 265 needed) |
| Tiertime UP600, Fab Lab E4-02-06 | 400 x 500 x 600 mm | fits | fits straight |
| any hobby printer with a 190 mm bed or less | | does not fit: ask for the two-piece track with links | ask for the two-piece plate |

So: **no split, no links**, as long as the tracks go on the UP300 or UP600. Keep the two-piece version as a fallback only.

## 2. What is made where, and the one-line justification for the request form

The Professional Workshop (E3-01-06) is CNC mills, a turn-mill and wire EDM: metal parts, done by the staff from a drawing. The Fab Lab (E4-02-06) is the 3D printers and the laser cutter. Job requests go through the EDIC request form.

| Part | Qty | Make how | Where | Justification (copy this) |
|---|---|---|---|---|
| Track | 4 | 3D print, TPU 95A, standing, 100% infill, 0.2 mm, no supports. File `stl/track_TPU_x4.stl` | Fab Lab UP300 (TPU head) | A toothed flexible belt with step-matched cleats on the outside; no stock belt has the cleats, and the cleat pitch (12.22 mm) is set by the ES1 step size. |
| Side plate, front L and R, rear L and R | 4 | **Option A:** 3D print PETG, flat, 4 walls, 30% infill. Files `stl/side_plate_{front,rear}_{L,R}.stl`. **Option B:** CNC-mill from 7 mm aluminium plate to sheets TS-02 / TS-03 (the drawings are made for this). | A: Fab Lab UP600 or UP300. B: Professional Workshop, by request form | One part carries the two bearing pockets, the slide rail under the lower track run and all holes; a cut sheet would need the rail and bearing housings as extra parts. |
| Cross bar spacer, 82 (front) and 65 (rear) | 2 + 4 | 3D print PETG (`stl/spacer_front_x2.stl`, `stl/spacer_rear_x4.stl`), or cut from Ø10 aluminium tube with a 4.5 bore on the lathe | Fab Lab, or Training Workshop with the plates | Non-standard lengths set the plate spacing; the rear ones sit between the box wall and the plate. |
| Stand-in pulley, PLA, test only | 1 | 3D print PLA (`stl/stand_in_pulley_20T_PLA_test_only.stl`) | Fab Lab, Tinkering Corner or NUS Library | Fit test of the printed teeth before the bought pulleys arrive. |

If printing is the hassle: **Option B for the plates** moves the biggest print (4 x 5 h) to the Professional Workshop staff, who machine it from the TS-02/TS-03 drawings. The tracks cannot be avoided: they must be printed in TPU. Everything else is bought.

## 3. Every piece, and which segment it belongs to

The prototype has two track segments: the **front segment** (hinged, plates inboard) and the **rear segment** (fixed to the box, plates outboard). They share the middle axle.

## Front segment (sheet TS-00)

| Item | Piece | Qty | Note |
|---|---|---|---|
| 1 | Track | 2 | one each side |
| 2 | Pulley HTD 5M 20T | 4 | 2 each side: one on the middle axle, one on the front axle |
| 3 | Front side plate | 2 | 1 left + 1 right (mirror images, not the same part) |
| 6 | Bearing 608ZZ | 4 | 2 pressed into each plate |
| 5 | Front axle Ø8 x 160 | 1 | |
| 7 | Spacer Ø10 x 82 | 2 | one per cross bar |
| 8 | M4 threaded rod x 110 | 2 | cut from the 1 m rod |
| 8 | M4 nut + M4 washer | 4 + 4 | one each end of each rod |
| 9 | M3 x 14 bolt + M3 nut | 6 + 6 | 3 per plate, hold the hinge sector gears (team's part) |
| - | M4 set screw | 8 | 2 per pulley; MISUMI does not include them |

## Rear segment

| Item | Piece | Qty | Note |
|---|---|---|---|
| 1 | Track | 2 | |
| 2 | Pulley HTD 5M 20T | 4 | 2 each side: middle axle and rear axle |
| - | Rear side plate | 2 | 1 left + 1 right |
| 6 | Bearing 608ZZ | 4 | 2 per plate |
| - | Rear axle Ø8 x 230 | 1 | |
| - | Spacer Ø10 x 65 | 4 | two per rod, one each side between the box wall (z 40) and the plate (z 105) |
| - | M4 threaded rod x 238 | 2 | **these are what bolt the segment to the box**: plate - spacer - box wall - box wall - spacer - plate (sheet TS-08) |
| - | M4 nut + washer | 4 + 4 | one each end of each rod |
| - | M4 set screw | 8 | |

## Shared

| Piece | Qty | Note |
|---|---|---|
| Middle axle Ø8 x 230 (item 4) | 1 | the hinge; carries 4 pulleys (2 front-segment, 2 rear-segment) and runs in all 4 plates' |

| | | |
|---|---|---|
| | | bearings |
| Hinge cross shaft Ø8 x 120 | 1 | runs across the box, carries the hinge pinions (team's part) |
| Bearing 608ZZ | 2 | in the box walls, for the cross shaft |

# Totals to get

| Piece | Total | Make / buy |
|---|---|---|
| Track | 4 | make |
| Pulley 20T | 8 | buy |
| Side plate | 4 (2 front L/R + 2 rear L/R) | make |
| Bearing 608ZZ | 10 | buy (2 packs of 5) |
| Ø8 rod, cut to 230 + 230 + 160 + 120 = 740 mm | 2 x 400 mm | buy (or Design Lab spare) |
| Spacer | 6 (2 x 82, 4 x 65) | make |
| M4 threaded rod, cut to 110 + 110 + 238 + 238 = 696 mm | 1 x 1 m | buy |
| M4 nut / M4 washer | 8 / 8 | buy or Design Lab spares |
| M3 x 14 bolt + nut | 6 + 6 | buy or spares |
| M4 set screw (M4 x 4 or x 5, cup point) | 16 | buy with the pulleys |
| Stand-in pulley, PLA | 1 | make |

## 4. How it goes together (one segment; the other is the same)

1. **Bearings into plates.** Press two 608ZZ into each plate's Ø22.2 pockets (a vice with wood packing, or a 22 mm socket as a drift). Flush with the plate face.
2. **Pulleys onto the axles.** Slide each pulley onto its axle, hub facing outboard (away from the plate), and set-screw it on the filed flat at the z station on sheet TS-04. On the middle axle the order from the centre out, each side, is: front plate bearing (z 41-48), front-segment pulley (49-69, hub 69-76), rear-segment pulley (77-97, hub 97-104), rear plate bearing (105-112).
3. **Track on, then the second plate.** Fit the left plate onto the middle and front axles. Loop a track over the two left pulleys (TPU stretches a little; walk it on like a bicycle chain). Then slide the right plate on and loop the right track. Doing the track before the outer plate is what makes it possible.
4. **Cross bars.** Front segment: M4 rod through one plate, the 82 spacer, the other plate; washer and nut each end. Rear segment: the same rods also hold the box. Rod through the plate, a 65 spacer, through the box wall, across the box, out the other wall, the second 65 spacer, the other plate; washer and nut each end, tightened until plates, spacers and box are clamped as one. Two per segment, at x 60 and 200 from the rear axle, 28 above the axle line. The box needs two Ø4.5 holes through each wall at those positions (sheet TS-08).
5. **Check:** turn the middle axle by hand, both tracks run without clicking; the lower run rests on the slide rails.
6. **Hinge (team's parts).** Sector gears bolt to the inboard face of the front plates with the six M3 x 14 and nuts (7 plate + 3 gear + nut). The front segment is held to the box only by the middle axle and these gears, so it can swing; nothing else is bolted.

## 5. Buy list with links

| Buy | Spec | Where | Price | Qty | Arrives |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Pulley | HTD 5M, 20 T, 15 mm belt, Ø8 bore, flanged, set-screw hub (MISUMI economy family C-HTPA20H5M150, bore 8) | MISUMI SG, Timing Pulleys HTD 5M | ~S$20 each (quote) | 8 | ships 1 day from SG stock per the catalogue; allow 2-5 working days |
| Set screws | M4 x 4 cup point (MISUMI MSSU4-3 or equal) | same order, or Design Lab | ~S$0.20 | 16 | with the pulleys |
| Bearing | 608ZZ (608-2Z/C3), 8 x 22 x 7, shielded | RS SG pack of 5, 261-2603 | S$14.82 per pack | 2 packs | next working day (order before 3:30 pm), max 4 working days |
| Ø8 rod | 8 mm x 400 mm steel shaft; hardened, so grind the flats (Dremel), or buy soft rod from a hardware shop on the vendor list | Design Lab spare first; else RS SG linear shafts | S$37 each at RS; ~S$5 at a hardware shop | 2 | next working day / same day |
| M4 rod | stainless M4 x 1 m, DIN 975 | RS SG 200-9958 (pack of 5, S$49.69) or one rod from a hardware shop | S$10 per rod | 1 | 4 working days / same day |
| Nuts, washers, bolts | M4 nuts x 8, M4 washers x 8, M3 x 14 + nuts x 6 | Design Lab spares; else Element14 SG nuts and RS SG | S$0 to S$15 | 22 | 1-2 business days |
| TPU filament | TPU 95A, 1.75 mm, 1 kg (4 tracks use about 130 g) | Element14 SG, Multicomp Pro, or the Fab Lab's own stock | ~S$45 | 1 | 1-2 business days |

**Totals:** about S$160 of Singapore-priced parts (bearings, rod, M4 rod, nuts), about S$45 with Design Lab spares, plus the pulleys (~S$160, quote) and TPU (~S$45). Whole set roughly **S$300-370 ex GST**. RS: free delivery over S$150 ex GST on a private account, else S$25 handling.

**Order of events:** day 0 order everything and print the stand-in pulley + fit-test slice; day 1-2 fit test, print or request the plates; day 2-5 parts arrive, press bearings, cut rod; day 3-6 print tracks, assemble one segment, run it on the test stairs. The pulleys are the only item that can hold the build up, so order them first.

# Motion and drive

, and how they link to the motor and the box

All of this is in `parts_spec.json` (`layout_decision`); it is repeated here because the drawings show the parts one at a time.

- **Every pulley is set-screwed to its axle** (a filed flat under each set screw). The **axles turn**

in the 608ZZ bearings pressed into the plates; the plates themselves do not turn with the axle.

- **One motor drives everything.** The drive motor turns the **middle axle** through a sprocket that

sits on it between the two front plates, inside the box (the box ends at z = 40 each side; the front plates are at z = 41 to 48). Because the middle axle carries the rear pulley of the front segment and the front pulley of the rear segment, both segments' tracks are driven from it. The rear and front axles are idlers: they turn because the track turns their pulleys.

- **The middle axle is also the hinge.** The front segment pivots about it on its plate bearings.

The hinge sector gear bolts to the rear end of each front plate (3 x M3 on a 14 mm radius); its pinion sits on the **hinge cross shaft**, which runs across the box in the two spare 608ZZ. Turning the cross shaft turns the sector gears, so the front segment bends up or down about the middle axle while the tracks keep running. The sector

gear, pinion and hinge motor are outside this drawing set.

- **The rear segment is fixed to the box.** Its outboard plates (z = 101 to 108) carry 2 x M4 holes

at x = -60 and -200 from the middle axle, 22 mm above the axle line, for the box cross bars (4 x M4 x 12). The front segment's plates are tied to each other only by their two cross bars (M4 rod + 82 mm spacer) so the segment can swing.

- **The lower run of each track rides on the plate's slide rail** (top face 13.34 mm below the axle),

so the track cannot sag between the pulleys when it is on the steps.